MATHEMATICS

TRIGONOMETRIC EQUATION

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TRIGONOMETRIC EQUATION

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GENERAL SOLUTIONS OF TRIGONOMETRIC EQUATIONS

1. The most general solution of $\tan\theta = -1$, $\cos\theta = \frac{1}{\sqrt{2}}$ is :

(A) $n\pi + \frac{7\pi}{4}$ (B) $n\pi + (-1)^n \frac{7\pi}{4}$ (C) $2n\pi + \frac{7\pi}{4}$ (D) None of these

निम्न का व्यापक हल करें $\tan\theta = -1$, $\cos\theta = \frac{1}{\sqrt{2}}$:

(A) $n\pi + \frac{7\pi}{4}$ (B) $n\pi + (-1)^n \frac{7\pi}{4}$ (C) $2n\pi + \frac{7\pi}{4}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\tan\theta = -1 = \tan\left(-\frac{\pi}{4}\right)$; $\cos\theta = \frac{1}{\sqrt{2}} = \cos\frac{\pi}{4}$

$\Rightarrow \theta = n\pi - \frac{\pi}{4}$; $\theta = 2n\pi \pm \frac{\pi}{4}$

If $n = 0$ then $\theta = -\frac{\pi}{4}$; $\theta = \frac{9\pi}{4}, \frac{7\pi}{4}$

$n = 2$ then $\theta = \frac{7\pi}{4}$; $\theta = \frac{17\pi}{4}, \frac{15\pi}{4}$

$\vdots$

$\Rightarrow \theta = \ldots\ldots, -\frac{\pi}{4}, \frac{7\pi}{4}, \frac{15\pi}{4}, \ldots\ldots\ldots$

$\Rightarrow \theta = 2n\pi + \frac{7\pi}{4}; n \in I$

SOLVING TRIGONOMETRIC EQUATION TYPE 1 (FACTORISATION)

2. $\frac{\sin 3\theta}{2\cos 2\theta + 1} = \frac{1}{2}$, if:

यदि $\frac{\sin 3\theta}{2\cos 2\theta + 1} = \frac{1}{2}$ तो θ का मान ज्ञात कीजिए :

(A) $\theta = n\pi + \frac{\pi}{6}$ (B) $\theta = 2n\pi - \frac{\pi}{6}$ (C) $\theta = n\pi + (-1)^n \frac{\pi}{6}$ (D) $\theta = n\pi - \frac{\pi}{6}$

Ans. C

Sol. $\frac{\sin 3\theta}{2\cos 2\theta+1}=\frac{1}{2}\Rightarrow 2\sin 3\theta=2\cos 2\theta+1$

$\Rightarrow 2(3\sin\theta-4\sin^3\theta)=2(1-2\sin^2\theta)+1$

$\Rightarrow 6\sin\theta-8\sin^3\theta=2-4\sin^2\theta+1$

$\Rightarrow 8\sin^3\theta-4\sin^2\theta=6\sin\theta-3$

$\Rightarrow (2\sin\theta-1)(4\sin^2\theta-3)=0$

$\Rightarrow \sin\theta=\frac{1}{2}$ or $\sin^2\theta=\frac{3}{4}=\sin^2\left(\frac{\pi}{3}\right)$

$\Rightarrow \theta=n\pi+(-1)^n\frac{\pi}{6}$ or $\theta=n\pi\pm\frac{\pi}{3}$

3. Find no. of roots of the equation $\tan x+\sec x=2\cos x$ in the interval $[0, 2\pi]$:

$[0, 2\pi]$ में समीकरण $\tan x+\sec x=2\cos x$ के मूलों की संख्या होगी :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\tan x+\sec x=2\cos x$; $x\in[0, 2\pi]$

$\Rightarrow \sin x+1=2\cos^2 x$; $x\neq\frac{\pi}{2},\frac{3\pi}{2}$

$\Rightarrow \sin x=-1$ or $\sin x=\frac{1}{2}$

In $0\le x\le 2\pi$

$\sin x=-1\Rightarrow x=\frac{3\pi}{2}$

& $\sin x=\frac{1}{2}\Rightarrow x=\frac{\pi}{6},\frac{5\pi}{6}$

$\Rightarrow$ No. of roots is 2 as tan x is not defined at $x=\frac{3\pi}{2}$

4. The number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2\sin^2 x+5\sin x-3=0$ is :

अन्तराल $[0, 3\pi]$ में x के उन मानों की संख्या जो समीकरण $2\sin^2 x+5\sin x-3=0$ को सन्तुष्ट करते है :

(A) 6 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $2\sin^2 x+5\sin x-3=0$; $x\in[0, 3\pi]$

$\Rightarrow \sin x=-3$ or $\sin x=\frac{1}{2}$

$\because \sin x \in [-1,1]$

$\Rightarrow \sin = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$

$\Rightarrow$ No. of values of x is 4.

SOLVING TRIGONOMETRIC EQUATION TYPE 2 (a sin x + b cos x = c)

5. If $0 < x < \pi$, and $\cos x + \sin x = \frac{1}{2}$, then tan x is :

यदि $0 < x < \pi$, तथा $\cos x + \sin x = \frac{1}{2}$, तो tan x का मान है :

(A) $(4 - \sqrt{7})/3$ (B) $-(4 + \sqrt{7})/3$ (C) $(1 + \sqrt{7})/4$ (D) $(1 - \sqrt{7})/4$

Ans. B

Sol. $0 < x < \pi$ and $\cos x + \sin x = 1/2$

$2(\cos x + \sin x) = 1$

Square both sides

$\Rightarrow 4[\cos^2 x + \sin^2 x + 2\sin x \cos x] = 1$

$\Rightarrow 4[1 + \tan^2 x + 2\tan x] = \sec^2 x$

$\Rightarrow 4[1 + \tan^2 x + 2\tan x] = 1 + \tan^2 x$

$\Rightarrow 4 + 4\tan^2 x + 8\tan x = 1 + \tan^2 x$

$\Rightarrow 3\tan^2 x + 8\tan x + 3 = 0$

$$\tan x = \frac{-8 \pm \sqrt{64 - 4.3.3}}{2.3} = \frac{-8 \pm 2\sqrt{7}}{2.3} = \frac{-4 \pm \sqrt{7}}{3}$$

6. Let $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta)$ and $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta)$ be two sets. Then :

माना $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta)$ तथा $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta)$ दो समुच्चय है तो :

(A) $P \subset Q$ and $Q - P \neq \phi$ (B) $Q \not\subset P$

(C) $P \not\subset Q$ (D) $P = Q$

Ans. D

Sol. $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$

$\Rightarrow \tan\theta = \sqrt{2} + 1$ ...(i)

$Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$ ...(ii)

from (i) & (ii)

$\Rightarrow$ P=Q

SOLVING TRIGONOMETRIC EQUATION TYPE 3 (SUM TO PRODUCT AND PRODUCT TO SUM)

7. The general solution of, $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ is :

$\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ का व्यापक हल है :

(A) $n\pi + \frac{\pi}{8}$ (B) $\frac{n\pi}{2} + \frac{\pi}{8}$ (C) $(-1)^n\left(\frac{n\pi}{2}\right) + \frac{\pi}{8}$ (D) $2n\pi + \cos^{-1}\left(\frac{3}{2}\right)$

Ans. B

Sol. $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$

$\Rightarrow 2\sin 2x\cos x - 3\sin 2x = 2\cos 2x\cos x - 3\cos 2x$

$\Rightarrow (\sin 2x - \cos 2x)(2\cos x - 3) = 0$

$\Rightarrow \sin 2x = \cos 2x$ or $\cos x = \frac{3}{2}$

$\Rightarrow \tan 2x = 1$ or $\underbrace{\cos x = \frac{3}{2} > 1}_{\text{Not possible as } \cos x \in [-1,1]}$

$\Rightarrow 2x = n\pi + \frac{\pi}{4}$

$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}; n \in I$

8. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are : **[AIEEE-2011]**

$\in (0, \pi)$ के ऐसे संभव मान कि $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ है, हैं : **[AIEEE-2011]**

(A) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$ (B) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(C) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$ (D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

Ans. A

Sol $\sin\theta + \sin 4\theta + \sin 7\theta = 0$; $\theta \in (0, \pi)$

$2\sin\left(\frac{\theta + 7\theta}{2}\right)\cos\left(\frac{7\theta - \theta}{2}\right) + \sin 4\theta = 0$

$\Rightarrow \sin 4\theta[2\cos 3\theta + 1] = 0$

$\Rightarrow \sin 4\theta = 0 \Rightarrow 4\theta = 0, \pi, 2\pi, 3\pi, 4\pi$

$\Rightarrow \theta = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$

but 0 and π are not included.

and $2\cos3\theta + 1 = 0 \Rightarrow \cos3\theta = \frac{-1}{2}$

$\Rightarrow 3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}$

$\Rightarrow \theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \frac{10\pi}{9}$

but $\frac{10\pi}{9} \notin (0, \pi)$

So, $\theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$

9. If $0 \le x < 2\pi$, then the number of real values of x, which satisfy the equation $\cos x + \cos2x + \cos3x + \cos4x = 0$, is : **[JEE(Main)2016]**

यदि $0 \le x < 2\pi$ है, तो x के उन वास्तविक मानों की संख्या जो समीकरण $\cos x + \cos2x + \cos3x + \cos4x = 0$ को सन्तुष्ट करते हैं, है : **[JEE(Main)2016]**

(A) 7 (B) 9 (C) 3 (D) 5

Ans. A

Sol Given $x \in [0, 2\pi)$

$\cos x + \cos2x + \cos3x + \cos4x = 0$

$\Rightarrow 2\cos\frac{5x}{2}\left(\cos\frac{3x}{2} + \cos\frac{x}{2}\right) = 0$

$\Rightarrow 4\cos\frac{5x}{2}\cos x \ \cos\frac{x}{2} = 0$

$\Rightarrow \cos\frac{5x}{2} = 0$ or $\cos x = 0$ or $\cos\frac{x}{2} = 0$

If $\cos\frac{5x}{2} = 0$ then

$x = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$

if $\cos x = 0$ then

$x = \frac{\pi}{2}, \frac{3\pi}{2}$

if $\cos\frac{x}{2} = 0$ then

$x = \pi$

There are total real values of x is 7.

10. If sum of all the solution of the equation $8\cos x.\left(\cos\left(\frac{\pi}{6}+x\right).\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right)=1$ in $[0, \pi]$ is $k\pi$, then k is equal to : **[JEE(Main)2018]**

यदि समीकरण $8\cos x.\left(\cos\left(\frac{\pi}{6}+x\right).\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right)=1$ के अंतराल $[0, \pi]$ में सभी हलों का योग $k\pi$ है, तो k बराबर है : **[JEE(Main)2018]**

(A) $\frac{8}{9}$ (B) $\frac{20}{9}$ (C) $\frac{2}{3}$ (D) $\frac{13}{9}$

Ans. D

Sol. $8\cos x.\cos\left(\frac{\pi}{6}+x\right)\cos\left(\frac{\pi}{6}-x\right)-4\cos x=1$

$\Rightarrow 8\cos x\left(\frac{3}{4}-\sin^2 x\right)-4\cos x=1$

$\Rightarrow 2\cos x\left(\frac{4\cos^2 x-1}{4}\right)-4\cos x=1$

$\Rightarrow 8\cos^3 x-2\cos x-4\cos x=1$

$\Rightarrow 8\cos^3x-6\cos x=1$

$\Rightarrow \cos3x=1/2$

$\Rightarrow 3x=2n\pi\pm\frac{\pi}{3}$

Given $x\in[0, \pi]$

$\Rightarrow x=\frac{\pi}{9},\frac{5\pi}{9},\frac{7\pi}{9}$

$\text{Sum}=\frac{13\pi}{9}$

11. If $5(\tan^2x-\cos^2x)=2\cos 2x+9$, then the value of $\cos 4x$ is : **[JEE(Main)2017]**

यदि $5(\tan^2x-\cos^2x)=2\cos 2x+9$ है, तो $\cos 4x$ का मान है : **[JEE(Main)2017]**

(A) $\frac{2}{9}$ (B) $-\frac{7}{9}$ (C) $-\frac{3}{5}$ (D) $\frac{1}{3}$

Ans. B

Sol. $5\left(\tan^2 x-\cos^2 x\right)=2\cos 2x+9$

$\Rightarrow 5\tan^2 x-5\cos^2 x=2\cos 2x+9$

$$\Rightarrow 5\left(\frac{1-\cos 2x}{1+\cos 2x}\right)-5\left(\frac{1+\cos 2x}{2}\right)=2\cos 2x+9$$

$$\Rightarrow 10(1-\cos 2x)-5(1+\cos 2x)^2=2(1+\cos 2x)(2\cos 2x+9)$$

$$\Rightarrow 10-10\cos 2x-5(1+\cos^2 2x+2\cos 2x)$$

$$=2[2\cos 2x+9+2\cos^2 2x+9\cos 2x]$$

$$= 5 - 20\cos 2x - 5\cos^2 2x = 4\cos 2x + 18 + 18\cos 2x + 4\cos^2 2x$$

$$\Rightarrow 9\cos^2 2x+42\cos 2x+13=0$$

$$\Rightarrow\left(\cos 2x+\frac{1}{3}\right)(9\cos 2x+39)=0$$

$$\Rightarrow \cos 2x=-\frac{1}{3} \quad \text{or} \quad \cos 2x=\frac{-39}{9}=\frac{-13}{3}<-1$$

$$\Rightarrow \cos 2x=-\frac{1}{3}$$

$$\Rightarrow \cos 4x=2\cos^2 2x-1=\frac{2}{9}-1=-\frac{7}{9}$$

SOLVING TRIGONOMETRIC EQUATION TYPE 5 (MAXIMUM AND MINIMUM BASED)

12. If $\theta \in [0, 5\pi]$ and $r \in R$ such that $2\sin\theta = r^4 - 2r^2 + 3$ then the maximum number of values of the pair (r, θ) is :
(A) 8 (B) 10 (C) 6 (D) none of these

यदि $\theta \in [0, 5\pi]$ और $r \in R$ इस प्रकार है कि $2\sin\theta = r^4 - 2r^2 + 3$ तब हल (r, θ) कि संख्या ज्ञात करें।
(A) 8 (B) 10 (C) 6 (D) इनमें से कोई नहीं

Ans. C

Sol. $\theta \in [0, 5\pi]; r \in R$

$2\sin\theta = r^4 - 2r^2 + 3$

$\Rightarrow 2\sin\theta = (r^2 - 1)^2 + 2$

$\because 2\sin\theta \in [-2, 2]$ & $(r^2 - 1)^2 + 2 \in [2, \infty)$

$\Rightarrow$ It is possible if

$2\sin\theta = 2$ & $(r^2 - 1)^2 + 2 = 2$

$\Rightarrow \sin\theta = 1$ & $(r^2 - 1)^2 = 0$

$\Rightarrow \theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}$ & $r = \pm 1$

$\Rightarrow$ Maximum number of values of the pair (r, θ) is 6.

13. The number of real solutions of, $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$ is :
(A) Zero (B) One (C) Two (D) Infinite

समीकरण $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$ के वास्तविक हलों की संख्या होगी :

(A) शून्य (B) एक (C) दो (D) अन्नत

Ans. A

Sol. $\sin e^x \cdot \cos e^x = 2^{x-2} + 2^{-x-2}$

$\Rightarrow \sin e^x \cdot \cos e^x = \frac{2^x}{4} + \frac{2^{-x}}{4}$

$\Rightarrow 4 \sin e^x + \cos e^x = 2^x + 2^{-x}$

$\Rightarrow 2 \sin (2e^x) = 2^x + 2^{-x}$

$\because 2 \sin (2e^x) \in [-2,2]$

& $2^x + 2^{-x} \in [2, \infty)$

$\Rightarrow$ This is possible if

$2 \sin (2e^x) = 2$ & $2^x + 2^{-x} = 2$

$\Rightarrow \underbrace{\sin\left(2e^x\right) = 1}_{\text{This is not possible if } x=0}$ & $x = 0$

$\Rightarrow$ No. of solution is zero.

14. The general solution of the equation , $\cos x \cos 6x = -1$ is :

(A) $x = (2n + 1)\pi, n \in I$ (B) $x = 2n\pi, n \in I$

(C) $x = (4n - 1)\pi, n \in I$ (D) None

समीकरण $\cos x \cos 6x = -1$ का व्यापक हल होगा :

(A) $x = (2n + 1)\pi, n \in I$ (B) $x = 2n\pi, n \in I$

(C) $x = (4n - 1)\pi, n \in I$ (D) इनमें से कोईनहीं

Ans. A

Sol. $\cos x \cos 6x = -1$

$\Rightarrow 2 \cos x \cos 6x = -2$

$\Rightarrow \cos 7x + \cos 5x = -2$

$\because \cos x \in [-1,1]$

$\Rightarrow \cos 7x = -1$ & $\cos 5x = -1$

$\Rightarrow 7x = (2n + 1)\pi; n \in I$ & $5x = (2n + 1)\pi; n \in I$

$\Rightarrow x = \left(\frac{2n+1}{7}\right)\pi; n \in I$ & $x = \left(\frac{2n+1}{5}\right)\pi; n \in I$

$\Rightarrow x = \pm\pi, \pm 2\pi, \pm 3\pi, ...$

$\Rightarrow x = (2n + 1)\pi; n \in I$

15. The number of solutions of the equation , $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$ is :

(A) Zero (B) Only one (C) Only two (D) Greater than 2

समीकरण $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}\,x + 4$ के हलों की संख्या है :

(A) शून्य (B) केवल एक (C) केवल दो (D) दो से बडा

Ans. B

Sol. $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$

$\because x^2 - 2\sqrt{3}x + 4 \in [1, \infty)$

& $\sin\left(\frac{\pi x}{2\sqrt{3}}\right) \in [-1, 1]$

$\Rightarrow$ This is possible if

$\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = 1$ & $x^2 - 2\sqrt{3}x + 4 = 1$

$\Rightarrow \frac{\pi x}{2\sqrt{3}} = (4n+1)\frac{\pi}{2}; n \in I$ & $\left(x - \sqrt{3}\right)^2 = 0$

$x = \sqrt{3}(4n+1); n \in I$ & $x = \sqrt{3}$

$\Rightarrow$ No. of solution is 1.

16. General solution for θ if, $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$ is :

(A) $2n\pi + \frac{7\pi}{6}$ (B) $2n\pi + \frac{\pi}{6}$ (C) $2n\pi - \frac{7\pi}{6}$ (D) None of these

समीकरण $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$ के लिए θ का व्यापक हल है :

(A) $2n\pi + \frac{7\pi}{6}$ (B) $2n\pi + \frac{\pi}{6}$ (C) $2n\pi - \frac{7\pi}{6}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\sin\left(2\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{5\pi}{6}\right) = 2$

$\because \sin\left(2\theta + \frac{\pi}{6}\right) \in [-1, 1]$

& $\cos\left(\theta + \frac{5\pi}{6}\right) \in [-1, 1]$

This is only possible if

$$\sin\left(2\theta+\frac{\pi}{6}\right)=1 \ \& \ \cos\left(\theta+\frac{5\pi}{6}\right)=1$$

$$\Rightarrow 2\theta+\frac{\pi}{6}=(4n+1)\frac{\pi}{2};n\in I \ \& \ \theta+\frac{5\pi}{6}=2n\pi;n\in I$$

$$\Rightarrow 2\theta=2n\pi+\frac{\pi}{3};n\in I \ \& \ \theta=2n\pi-\frac{5\pi}{6};n\in I$$

$$\Rightarrow \theta=n\pi+\frac{\pi}{6};n\in I \ \& \ \theta=2n\pi-\frac{5\pi}{6};n\in I$$

$\Rightarrow$ General solution for θ is

$$2n\pi+\frac{\pi}{6};n\in I$$

17. The number of values of 'x' for which $\sin 2x+\cos 4x=2$ is :

(A) 0 (B) 1 (C) 2 (D) infinite

'x' के मानों की संख्या ज्ञात करें जो समीकरण $\sin 2x+\cos 4x=2$ को संतुष्ट करते है :

(A) 0 (B) 1 (C) 2 (D) अन्नत

Ans. A

Sol. $\sin 2x+\cos 4x=2$

$\because \sin 2x\in[-1,1] \ \& \ \cos 4x\in[-1,1]$

$\Rightarrow$ This is only possible if

$\sin 2x=1 \ \& \ \cos 4x=1$

$$\Rightarrow 2x=(4n+1)\frac{\pi}{2};n\in I \ \& \ 4x=2n\pi;n\in I$$

$$\Rightarrow x=n\pi+\frac{\pi}{4};n\in I \ \& \ x=\frac{n\pi}{2};n\in I$$

$$\Rightarrow x=\frac{\pi}{4},\frac{5\pi}{4},\frac{9\pi}{4},..... \ \& \ x=0,\frac{\pi}{2},\pi,\frac{3\pi}{2},.....$$

$\Rightarrow$ No. of value of x is zero.

18. The equation , $4\sin^2 x+4\sin x+a^2-3=0$ possesses a solution if 'a' belongs to the interval :

यदि समीकरण $4\sin^2 x+4\sin x+a^2-3=0$ का हल संभव होगा यदि 'a' निम्न अन्तराल में होगा :

(A) $(-1,3)$ (B) $(-3,1)$ (C) $[-2,2]$ (D) $R-(-2,2)$

Ans. C

Sol. $4\sin^2x+4\sin x+a^2-3=0$

$\Rightarrow 4\sin^2x+4\sin x=3-a^2$

$\Rightarrow 4[\sin^2x+\sin x]=3-a^2$

$$\Rightarrow\left[\left(\sin x+\frac{1}{2}\right)^2\right]=4-a^2$$

$\because \left(\sin x+\frac{1}{2}\right)^2 \in [0,1]$

$\Rightarrow 4\left(\sin x+\frac{1}{2}\right)^2 \in [0,4]$

$\Rightarrow 4 - a^2 \in [0, 4]$

$\Rightarrow 0 \le 4 - a^2 \le 4$

$\Rightarrow -4 \le -a^2 \le 0$

$\Rightarrow 0 \le a^2 \le 4$

$\Rightarrow a \in [-2, 2]$

19. The equation, $\cos 2x + a \sin x = 2a - 7$ possesses a solution if :

(A) $a < 2$ (B) $2 \le a \le + 6$ (C) $a > 6$ (D) a is any integer

समीकरण $\cos 2x + a \sin x = 2a - 7$ का हल विद्यमान है, यदि :

(A) $a < 2$ (B) $2 \le a \le + 6$ (C) $a > 6$ (D) a कोई भी पूर्णांक है।

Ans. B

Sol. $\cos 2x + a \sin x = 2a - 7$

i.e. $2\sin^2 x - a \sin x + 2a - 8 = 0$

$$\sin x = \frac{a \pm \sqrt{a^2 - 8(2a-8)}}{4} = \frac{a \pm (a-8)}{4}$$

$$\sin x = \frac{a-4}{2} \text{ or } \underbrace{\sin x = 2}_{\text{not possible}}$$

Hence $-1 \le (a-4)/2 \le 1 \quad \Rightarrow \quad a \in [2, 6]$

TRIGONOMETRIC INEQUALITIES

20. If $0 < \theta < 2\pi$, then the intervals of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$, is :

यदि $0 < \theta < 2\pi$ तो θ का अन्तराल जिनके लिये $2\sin^2\theta - 5\sin\theta + 2 > 0$ है :

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$

(C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{41\pi}{48}, \pi\right)$

Ans. A

Sol. $2\sin^2\theta - 5\sin\theta + 2 > 0$

Given $\theta \in (0, 2\pi)$

$\Rightarrow 2\sin 2\theta - 4\sin\theta - \sin\theta + 2 > 0$

$\Rightarrow (\sin\theta - 2)(2\sin\theta - 1) > 0$

$\because \sin\theta \in [-1, 1]$

$\Rightarrow \sin\theta - 2 < 0$

$\Rightarrow 2\sin\theta - 1 < 0$

$\Rightarrow \sin\theta < \frac{1}{2}$

$\because \theta \in (0, 2\pi)$

$\Rightarrow \theta \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$

SYSTEM OF TRIGONOMETRIC EQUATION

21. $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = 1/e$, where $\alpha, \beta \in [-\pi, \pi]$, numbers of pairs of α, β which satisfy both the equations is :

$\cos(\alpha - \beta) = 1$ तथा $\cos(\alpha + \beta) = 1/e$, जहाँ $\alpha, \beta \in [-\pi, \pi]$, ऐसे कितने α, β के युग्म होगें जो दोनों समीकरणों को संतुष्ट होगें :

(A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $\cos(\alpha - \beta) = 1$ & $\cos(\alpha + \beta) = \frac{1}{e}$

Given $\alpha \in [-\pi, \pi]$; $\beta \in [-\pi, \pi]$

$\Rightarrow \alpha + \beta \in [-2\pi, 2\pi]$; $\alpha - \beta \in [-2\pi, 2\pi]$

$\because \cos(\alpha - \beta) = 1$

$\Rightarrow \alpha - \beta = -2\pi, 0, 2\pi$

$\Rightarrow (\alpha, \beta) = (-\pi, \pi), (\alpha, \alpha), (\pi, -\pi)$

For $\cos(\alpha + \beta) = \frac{1}{e}$; rejected value of $(\alpha, \beta) = (-\pi, \pi), (\pi, -\pi)$

$\Rightarrow \cos 2\alpha = \frac{1}{e}$

$\because 2\alpha \in [-2\pi, 2\pi]$ & $\frac{1}{e} < 1$

$\Rightarrow$ No. of pairs of α, β are 4.

22. The number of solutions of the pair of equations $2\sin^2\theta - \cos 2\theta = 0$ $2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is : **[JEE-2007]**

(A) zero (B) one (C) two (D) four

समीकरणों $2\sin^2\theta - \cos 2\theta = 0$, $2\cos^2\theta - 3\sin\theta = 0$ के अन्तराल $[0, 2\pi]$ में हलों की संख्या है : **[JEE-2007]**

(A) शन्यू (B) एक (C) दो (D) चार

Ans. C

Sol. $2\sin^2\theta - \cos 2\theta = 0$, $2\cos^2\theta - 3\sin\theta = 0$

$\Rightarrow 2\sin^2\theta - (1 - 2\sin^2\theta) = 0$; $2(1 - \sin^2\theta) - 3\sin\theta = 0$

$\Rightarrow \sin^2\theta = \frac{1}{4}$; $2\sin^2\theta + 3\sin\theta - 2 = 0$

$\Rightarrow \sin^2\theta = \frac{1}{4}$; $\sin\theta = \frac{1}{2}$

$\Rightarrow$ Number of solution of the pair of equation $2\sin^2\theta - \cos 2\theta = 0$, $2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is 2.

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

SOLVING TRIGONOMETRIC EQUATION TYPE 1 (FACTORISATION)

1. Find the general values of θ for which the quadratic function $(\sin\theta)\,x^2 + (2\cos\theta)x + \frac{\cos\theta+\sin\theta}{2}$ is the square of a linear function.

θ के वह व्यापक मान ज्ञात कीजिए जिनके जिनके लिए द्विघात फलन $(\sin\theta)\,x^2 + (2\cos\theta)x + \frac{\cos\theta+\sin\theta}{2}$ एक रेखीय फलन का वर्ग है।

Ans. $2n\pi+\frac{\pi}{4}$ or $(2n+1)\pi - \tan^{-1}2$, $n \in I$

Sol. Given quadratic function

$\left(\sin\theta\right)x^2 + \left(2\cos\theta\right)x + \frac{\cos\theta+\sin\theta}{2}$ is the square of a linear function.

$\Rightarrow \sin\theta > 0$ & $D \equiv 4\cos^2\theta - 4\sin\theta\left(\frac{\sin\theta+\cos\theta}{2}\right) = 0$

$\Rightarrow \sin\theta > 0$ & $4\cos^2\theta - 2\sin^2\theta - 2\sin\theta\cos\theta = 0$

$\Rightarrow \sin\theta > 0$ & $(4\cos\theta + 2\sin\theta)(\cos\theta - \sin\theta) = 0$

$\Rightarrow \sin\theta > 0$ & $(\cos\theta - \sin\theta)(\sin\theta + 2\cos\theta) = 0$

$\Rightarrow \sin\theta > 0$ & $\tan\theta = 1$ or $\tan\theta = -2$

$\Rightarrow \sin\theta > 0$ & $\tan\theta = 1$ or $\sin\theta > 0$ & $\tan\theta = -2$

$\Rightarrow \theta = 2n\pi + \frac{\pi}{4}; n \in I$ or $\theta = \left(2n+1\right)\pi - \tan^{-1}\left(2\right); n \in I$

2. Solve the equality: $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$.

समीकरण : $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$ को हल कीजिए।

Ans. $x = \frac{n\pi}{7} - \frac{\pi}{84}$ or $x = \frac{n\pi}{4} + \frac{7\pi}{48}$, $n \in I$

Sol. $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$

$\Rightarrow 2\sin 11x + 2\left(\frac{1}{2}\cos 3x + \frac{\sqrt{3}}{2}\sin 3x\right) = 0$

$\Rightarrow 2\sin 11x + 2\sin\left(3x + \frac{\pi}{6}\right) = 0$

$$\Rightarrow \sin 11x + \sin\left(3x + \frac{\pi}{6}\right) = 0$$

$$\Rightarrow 2\sin\left(\frac{11x + 3x + \frac{\pi}{6}}{2}\right)\cos\left(\frac{11x - 3x - \frac{\pi}{6}}{2}\right) = 0$$

$$\Rightarrow \sin\left(\frac{14x + \frac{\pi}{6}}{2}\right) = 0 \quad \text{or} \quad \cos\left(\frac{8x - \frac{\pi}{6}}{2}\right) = 0$$

$$\Rightarrow \frac{14x + \frac{\pi}{6}}{2} = n\pi; n \in I \quad \text{or} \quad \frac{8x - \frac{\pi}{6}}{2} = (2n+1)\frac{\pi}{2}; n \in I$$

$$\Rightarrow x = \frac{n\pi}{7} - \frac{\pi}{84}; n \in I \quad \text{or} \quad x = \frac{n\pi}{4} + \frac{7\pi}{48}; n \in I$$

SOLVING TRIGONOMETRIC EQUATION TYPE 3 (SUM TO PRODUCT AND PRODUCT TO SUM)

3. Find all the values of θ satisfying the equation; $\sin\theta + \sin 5\theta = \sin 3\theta$ such that $0 \le \theta \le \pi$.

θ के सभी मान ज्ञात कीजिए जो समीकरण $\sin\theta + \sin 5\theta = \sin 3\theta$ को संतुष्ट करते है, जहाँ $0 \le \theta \le \pi$.

Ans. $0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}$ and π

Sol. $\sin\theta + \sin 5\theta = \sin 3\theta; \quad 0 \le \theta \le \pi$

$2\sin 3\theta \cos 2\theta - \sin 3\theta = 0$

$$\Rightarrow \sin 3\theta = 0 \quad \text{or} \quad \cos 2\theta = \frac{1}{2}$$

Given $0 \le \theta \le \pi$

$$\Rightarrow 3\theta = 0, \pi, 2\pi, 3\pi \quad \text{or} \quad 2\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\Rightarrow \theta = 0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}, \pi$$

4. Find all values of θ between $0°$ and $180°$ satisfying the equation; $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$.

अन्तराल $0°$ तथा $180°$ में समीकरण $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$ को संतुष्ट करने वाले θ के सभी मान ज्ञात कीजिए।

Ans. $30°, 45°, 90°, 135°, 150°$

Sol. Given

$\theta \in [0, \pi]$

$\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$

$\Rightarrow 2\cos 5\theta \cdot \cos\theta + 2\cos^2\theta = 0$

$\Rightarrow 2\cos\theta\,(\cos 5\theta + \cos\theta) = 0$

$\Rightarrow 2\cos\theta \cdot 2\cos 3\theta \cos 2\theta = 0$

$\Rightarrow \cos\theta \cdot \cos 2\theta \cdot \cos 3\theta = 0$

$\Rightarrow \cos\theta = 0$ or $\cos 2\theta = 0$ or $\cos 3\theta = 0$

$\Rightarrow \theta = 90^\circ$ or $2\theta = 90^\circ, 270^\circ$ or $3\theta = 90^\circ, 270^\circ, 450^\circ$

$\Rightarrow \theta = 30^\circ, 45^\circ, 90^\circ, 135^\circ, 150^\circ$

5. The positive integer value of $n > 3$ satisfying the equation $\dfrac{1}{\sin\left(\frac{\pi}{n}\right)} = \dfrac{1}{\sin\left(\frac{2\pi}{n}\right)} + \dfrac{1}{\sin\left(\frac{3\pi}{n}\right)}$ is.

समीकरण $\dfrac{1}{\sin\left(\frac{\pi}{n}\right)} = \dfrac{1}{\sin\left(\frac{2\pi}{n}\right)} + \dfrac{1}{\sin\left(\frac{3\pi}{n}\right)}$ को सन्तुष्ट करने वाले धनात्मक पूर्णांक $n(>3)$ का मान होगा।

Ans. 7

Sol. Given $\dfrac{1}{\sin\frac{\pi}{n}} = \dfrac{1}{\sin\frac{2\pi}{n}} + \dfrac{1}{\sin\frac{3\pi}{n}}$ & $n > 3$

$$\Rightarrow \frac{1}{\sin\frac{\pi}{n}} - \frac{1}{\sin\frac{3\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}}$$

$$\Rightarrow \frac{\sin\frac{3\pi}{n} - \sin\frac{\pi}{n}}{\sin\frac{\pi}{n}\sin\frac{3\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}}$$

$$\Rightarrow \frac{2\cos\frac{2\pi}{n}\sin\frac{\pi}{n}}{\sin\frac{\pi}{n}\sin\frac{3\pi}{n}} = \frac{1}{\sin\frac{2\pi}{n}}$$

$$\Rightarrow 2\cos\frac{2\pi}{n}\sin\frac{2\pi}{n} = \sin\frac{3\pi}{n}$$

$\Rightarrow \sin\frac{4\pi}{n} = \sin\frac{3\pi}{n} \Rightarrow \frac{4\pi}{n} = K\pi + (-1)^K \frac{3\pi}{n}$

If $K = 2m$ then $\frac{\pi}{n} = 2m\pi$

$\Rightarrow \quad n = \frac{1}{2m}$ $\Rightarrow n = \frac{1}{2}, \frac{1}{4}, \frac{1}{6}$ .....

If $K = 2m + 1$ then $\frac{7\pi}{n} = (2m+1)\pi$

$\Rightarrow \quad n = \frac{7}{2m+1}$ $\Rightarrow \quad n = 7, \frac{7}{3}, \frac{7}{5}$.....

Given $n > 3 \Rightarrow n = 7$

SOLVING TRIGONOMETRIC EQUATION TYPE 4 (SUBSTITUTION BASED)

6. Find all value of θ, between 0 & π, which satisfy the equation; $\cos\theta \cdot \cos 2\theta \cdot \cos 3\theta = 1/4$.

अन्तराल 0 व π के मध्य θ के सभी मान ज्ञात कीजिए जो समीकरण $\cos\theta \cdot \cos 2\theta \cdot \cos 3\theta = 1/4$ को संतुष्ट करते है।

Ans. $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$

Sol. $\cos\theta \cdot \cos 2\theta \cdot \cos 3\theta = \frac{1}{4}$

$\cos\theta(2\cos^2\theta - 1) \cdot (4\cos^3\theta - 3\cos\theta) = \frac{1}{4}$

$\Rightarrow (2\cos^3\theta - \cos\theta)(4\cos^3\theta - 3\cos\theta) = \frac{1}{4}$

$\Rightarrow 8\cos^6\theta - 6\cos^4\theta - 4\cos^4\theta + 3\cos^2\theta = \frac{1}{4}$

$\Rightarrow 8\cos^6\theta - 10\cos^4\theta + 3\cos^2\theta = \frac{1}{4}$

Let $\cos^2\theta = t$

$\Rightarrow 8t^3 - 10t^2 + 3t = \frac{1}{4}$

$\Rightarrow 32t^3 - 40t^2 + 12t - 1 = 0$

$\Rightarrow \left(t - \frac{1}{4}\right)\left(32t^2 - 32t + 4\right) = 0$

$\Rightarrow \left(t - \frac{1}{4}\right)\left(8t^2 - 8t + 1\right) = 0$

$\Rightarrow t = \frac{1}{4}$ or $t = \frac{\sqrt{2}+1}{2\sqrt{2}}$ or $t = \frac{\sqrt{2}-1}{2\sqrt{2}}$

$\Rightarrow \cos^2\theta = \frac{1}{4}$ or $\cos^2\theta = \frac{\sqrt{2}+1}{2\sqrt{2}}$ or $\cos^2\theta = \frac{\sqrt{2}-1}{2\sqrt{2}}$

$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}$ or $\theta = n\pi \pm \frac{\pi}{8}$ or $\theta = n\pi \pm \frac{3\pi}{8}$

Given $\theta \in (0, \pi)$

$\Rightarrow \theta = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \frac{\pi}{8}, \pi - \frac{\pi}{8}, \frac{3\pi}{8}, \pi - \frac{3\pi}{8}$

$\Rightarrow \theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$

7. Find the general solution of the equation , $2 + \tan x \cdot \cot \frac{x}{2} + \cot x \cdot \tan \frac{x}{2} = 0.$

समीकरण $2 + \tan x \cdot \cot \frac{x}{2} + \cot x \cdot \tan \frac{x}{2} = 0$ का व्यापक हल बताइए।

Ans. $x = 2n\pi \pm \frac{2\pi}{3}, n \in I$

Sol. $2 + \tan x \cdot \cot\frac{x}{2} + \cot x \cdot \tan\frac{x}{2} = 0$

$\Rightarrow 2 + \frac{\sin x}{\cos x} \cdot \frac{\cos\frac{x}{2}}{\sin\frac{x}{2}} + \frac{\cos x}{\sin x} \cdot \frac{\sin\frac{x}{2}}{\cos\frac{x}{2}} = 0$

$\Rightarrow 2 + \frac{2\cos^2\frac{x}{2}}{\cos x} + \frac{\cos x}{2\cos^2\frac{x}{2}} = 0$ $\left\{\begin{array}{l}\because \cos x \neq 0 \\ \& \ 1+\cos x \neq 0\end{array}\right\}$

$\Rightarrow 2 + \frac{1+\cos x}{\cos x} + \frac{\cos x}{1+\cos x} = 0$

$\Rightarrow 2(\cos x + \cos^2 x) + 1 + \cos^2 x + 2\cos x + \cos^2 x = 0$

$\Rightarrow 4\cos^2 x + 4\cos x + 1 = 0$

$\Rightarrow (2\cos x + 1)^2 = 0$

$\Rightarrow \cos x = -\frac{1}{2}$

$\Rightarrow x = 2n\pi \pm \frac{2\pi}{3}; n \in I$

8. Sum of all the solutions in $[0, 4\pi]$ of the equation $\tan x + \cot x + 1 = \cos\left(x + \frac{\pi}{4}\right)$ is $k\pi$ then the value of k is.

अन्तराल $[0, 4\pi]$ में समीकरण $\tan x + \cot x + 1 = \cos\left(x + \frac{\pi}{4}\right)$ के सभी हलों का योग $k\pi$ है, तो k का मान ज्ञात कीजिये।

Ans. 3.5

Sol. Given $x \in [0, 4\pi]$

$\tan x + \cot x + 1 = \cos\left(x + \frac{\pi}{4}\right)$

$\because \tan x + \cot x \in (-\infty, -2] \cup [2, \infty)$

$\Rightarrow \tan x + \cot x + 1 \in (-\infty, -1] \cup [3, \infty)$

& $\cos\left(x + \frac{\pi}{4}\right) \in [-1, 1]$

This is only possible only when

$\cos\left(x + \frac{\pi}{4}\right) = -1$ & $\tan x + \cot x + 1 = -1$

$\Rightarrow x + \frac{\pi}{4} = (2n+1)\pi$ & $\tan x + \cot x = -2$

$\Rightarrow x = (2n+1)\pi - \frac{\pi}{4}$ & $\tan^2 x + 2\tan x + 1 = 0$

& $(\tan x + 1)^2 = 0$

$\because x \in [0, 4\pi]$

$\Rightarrow x = \frac{3\pi}{4}, \frac{11\pi}{4}$ & $\tan x = -1$

$\Rightarrow x = \frac{3\pi}{4}, \frac{11\pi}{4}$ & $x = \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{11\pi}{4}$

$\Rightarrow x = \frac{3\pi}{4}, \frac{11\pi}{4}$

$\Rightarrow$ Sum of solutions $= \frac{3\pi}{4} + \frac{11\pi}{4} = \frac{14\pi}{4} = \frac{7\pi}{2}$

$\Rightarrow k = \frac{7}{2} = 3.5$

SOLVING TRIGONOMETRIC EQUATION TYPE 5 (MAXIMUM AND MINIMUM BASED)

9. Find the range of y such that the equation, $y + \cos x = \sin x$ has a real solution. For $y = 1$, find x such that $0 < x < 2\pi$.

y का परिसर ज्ञात कीजिए जिसके लिये समीकरण $y + \cos x = \sin x$ का कोई वास्तविक हल हो। $y = 1$ के लिये x का $(0 < x < 2\pi)$ में मान ज्ञात कीजिए।

Ans. $-\sqrt{2} \le y \le \sqrt{2}$; $\frac{\pi}{2}, \pi$

Sol. $y + \cos x = \sin x$

$\Rightarrow y = \sin x - \cos x$

$\Rightarrow$ Range of y is $\left[-\sqrt{2}, \sqrt{2}\right]$

If $y = 1$ then $1 + \cos x = \sin x$

$\Rightarrow \sin x - \cos x = 1$

$$\Rightarrow \frac{\sin x}{\sqrt{2}} - \frac{\cos x}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} = \cos\left(\frac{3\pi}{4}\right)$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi + \frac{3\pi}{4} \quad \text{or} \quad x + \frac{\pi}{4} = 2n\pi - \frac{3\pi}{4}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{2}; n \in I \quad \text{or} \quad x = 2n\pi - \pi; n \in I$$

$\because x \in (0, 2\pi)$

$$\Rightarrow x = \frac{\pi}{2} \quad \text{or} \quad x = \pi$$

$$\Rightarrow x = \frac{\pi}{2}, \pi$$

10. Let $f(x) = \sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$ for some real number k. Determine :

(a) all real numbers k for which $f(x)$ is constant for all values of x.

(b) all real numbers k for which there exists a real number 'c' such that $f(c) = 0$.

(c) If $k = -0.7$, determine all solutions to the equation $f(x) = 0$.

माना $f(x) = \sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$, जहाँ k कोई वास्तविक संख्या है, तो :

(a) k के वह सभी वास्तविक मान ज्ञात कीजिये जिनके लिये f(x), x के सभी मानों के लिये, अचर है।

(b) k के वह सभी वास्तविक मान ज्ञात कीजिये जिनके लिये एक वास्तविक संख्या 'c' इस प्रकार विद्यमान है कि $f(c) = 0$ हो।

(c) यदि $k = -0.7$ है, तो समीकरण $f(x) = 0$ के सभी हल ज्ञात कीजिये।

Ans. (a) $-\frac{3}{2}$; (b) $k \in \left[-1, -\frac{1}{2}\right]$; (c) $x = \frac{n\pi}{2} \pm \frac{\pi}{6}$

Sol. $f(x) = \sin^6 x + \cos^6 x + k(9\sin^4 x + \cos^4 x)$

$\Rightarrow f(x) = 1 - 3\sin^2 x \cos^2 x + k(1 - 2\sin^2 x \cos^2 x)$

$\Rightarrow f(x) = (k + 1) - \sin^2 x \cos^2 x (2k + 3)$

(a) Given all real numbers k for which f(x) is constant for all values of x

$\Rightarrow 2k + 3 = 0$

$$\Rightarrow k = -\frac{3}{2}$$

(b) Given all real numbers k for which there exists a real number c such that $f(c) = 0$

$\Rightarrow (k + 1) - (2k + 3)\sin^2 c \cos^2 c = f(c) = 0$

$$\Rightarrow \frac{\sin^2(2c)}{4} = \frac{k+1}{2k+3}$$

$\because \sin^2(2c) \in [0,1]$

$$\Rightarrow 0 \le \frac{4(k+1)}{2k+3} \le 1$$

$$0 \le \frac{4(k+1)}{2k+3} \quad \& \quad \frac{4k+4-2k-3}{2k+3} \le 0$$

$$\Rightarrow \frac{4k+4}{2k+3} \ge 0 \quad \& \quad \frac{2k+1}{2k+3} \le 0$$

$$\Rightarrow k \in \left(-\infty, -\frac{3}{2}\right) \cup [-1, \infty) \quad \& \quad k \in \left(-\frac{3}{2}, -\frac{1}{2}\right]$$

$$\Rightarrow k \in \left[-1, -\frac{1}{2}\right]$$

(c) If $k = -0.7 = \frac{-7}{10}$

then $f(x) = \left(\frac{-7}{10} + 1\right) - \sin^2 x \cos^2 x \left(\frac{-14}{10} + 3\right) = 0$

$$\Rightarrow \frac{3}{10} - \frac{16}{10}\sin^2 x \cos^2 x = 0$$

$$\Rightarrow \sin^2 2x = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2 = \sin^2 \frac{\pi}{3}$$

$\Rightarrow 2x = n\pi \pm \frac{\pi}{3}$

$\Rightarrow x = \frac{n\pi}{2} \pm \frac{\pi}{6}; n \in I$

SOLVING TRIGONOMETRIC EQUATION TYPE 6 (SQUARING BASED)

11. Solve the equation for x, $5^{\frac{1}{2}} + 5^{\frac{1}{2}+\log_5(\sin x)} = 15^{\frac{1}{2}+\log_{15}\cos x}$.

समीकरण $5^{\frac{1}{2}} + 5^{\frac{1}{2}+\log_5(\sin x)} = 15^{\frac{1}{2}+\log_{15}\cos x}$ को x के लिये हल कीजिए।

Ans. $x = 2n\pi + \frac{\pi}{6}, \ n \in I$

Sol. $5^{\frac{1}{2}} + 5^{\frac{1}{2}+\log_5(\sin x)} = 15^{\frac{1}{2}+\log_{15}\cos x}$

$\Rightarrow 5^{\frac{1}{2}} + 5^{\frac{1}{2}} \cdot \sin x = 15^{\frac{1}{2}} \cdot \cos x$

$\Rightarrow 1 + \sin x = \sqrt{3}\cos x$

$\Rightarrow \sqrt{3}\cos x - \sin x = 1$

$\Rightarrow \frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x = \frac{1}{2}$

$\Rightarrow \cos\left(x + \frac{\pi}{6}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

$\Rightarrow x + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$

$\Rightarrow x = 2n\pi + \frac{\pi}{6}$ or $x = 2n\pi - \frac{\pi}{2}$

$\because \sin x > 0 \ \& \ \cos x > 0$

$\Rightarrow x = 2n\pi + \frac{\pi}{6}; n \in I$

12. Solve for x, the equation $\sqrt{13 - 18\tan x} = 6\tan x - 3$, where $-2\pi < x < 2\pi$.

अन्तराल $-2\pi < x < 2\pi$ में समीकरण $\sqrt{13 - 18\tan x} = 6\tan x - 3$ के लिये x के मान निकालिए।

Ans. $\alpha - 2\pi$; $\alpha - \pi$, α, $\alpha + \pi$, where $\tan\alpha = \frac{2}{3}$

Sol. $\sqrt{13 - 18\tan x} = 6\tan x - 3; -2\pi < x < 2\pi$

square both sides; we get

$\Rightarrow 13 - 18\tan x = 36\tan^2 x + 9 - 36\tan x$

$\Rightarrow 36\tan^2 x - 18\tan x - 4 = 0$

$\Rightarrow 18\tan^2 x - 9\tan x - 2 = 0$

$\Rightarrow (3\tan x - 2)(6\tan x + 1) = 0$

$\Rightarrow \tan x = \frac{2}{3}$ or $\tan x = \frac{-1}{6}$

$\left\{\begin{array}{ll} \text{For } \sqrt{13-18\tan x} & 13-18\tan x \geq 0 \\ \Rightarrow \tan \leq \frac{13}{18} & \\ \tan x \geq \frac{1}{2} & \text{as } 6\tan x - 3 \geq 0 \end{array}\right\}$

$\Rightarrow \tan x = \frac{2}{3}$

$\Rightarrow x = n\pi + \alpha;$ $\alpha = \tan^{-1}\left(\frac{2}{3}\right)$

$\Rightarrow x = \alpha,\quad \pi + \alpha,\quad -\pi + \alpha,\quad -2\pi + \alpha$

13. Determine the smallest positive value of x which satisfy the equation, $\sqrt{1 + \sin 2x} - \sqrt{2}\ \cos 3x = 0$.

समीकरण $\sqrt{1 + \sin 2x} - \sqrt{2}\ \cos 3x = 0$ को संतुष्ट करने वाले x की न्यूनतम धनात्मक मान बताइए।

Ans. $x = \pi/16$

Sol. $\sqrt{1+\sin 2x} = \sqrt{2}\cos 3x$

If $\cos 3x \geq 0$

then

$1 + \sin 2x = 2\cos^2 3x$

$\Rightarrow \sin 2x = 2\cos^2 3x - 1$

$\Rightarrow \sin 2x = \cos 6x$

$\Rightarrow \cos 6x = \cos\left(\frac{\pi}{2} - 2x\right)$

$\Rightarrow 6x = 2n\pi \pm \left(\frac{\pi}{2} - 2x\right)$

$\Rightarrow 8x = 2n\pi + \frac{\pi}{2}$ or $4x = 2n\pi - \frac{\pi}{2}; n \in I$

$\Rightarrow x = \frac{n\pi}{4} + \frac{\pi}{16}$ or $x = \frac{n\pi}{2} - \frac{\pi}{8}; n \in I$

$\therefore$ Smallest positive value of x is $\frac{\pi}{16}$.

14. $2 \sin\left(3x + \frac{\pi}{4}\right) = \sqrt{1 + 8 \sin 2x \,.\, \cos^2 2x}$.

Ans. $x = 2n\pi + \frac{\pi}{12}$ or $2n\pi + \frac{17\pi}{12}$; $n \in I$

Sol. $2\sin(3x + \frac{\pi}{4}) = \sqrt{1 + 8\sin 2x \cos^2 2x}$

$\Rightarrow \; 2\sin\left(3x + \frac{\pi}{4}\right) = \sqrt{1 + 4\sin 4x \cos 2x}$

$\Rightarrow \; 4\sin^2\left(3x + \frac{\pi}{4}\right) = 1 + 4\sin 4x \cos 2x$. If $\sin\left(3x + \frac{\pi}{4}\right) \geq 0$

$\Rightarrow \; 2 - 2\cos\left(6x + \frac{\pi}{2}\right) = 1 + 2\sin 6x + 2\sin 2x$

$\Rightarrow 2 + 2\sin 6x = 1 + 2\sin 6x + 2\sin 2x$

$\Rightarrow \; \sin 2x = \frac{1}{2} \Rightarrow x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}, n \in I$

$\Rightarrow \; x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

$\because \; \sin\left(3x + \frac{\pi}{4}\right) \geq 0$ so $n = \frac{\pi}{12}, \frac{17\pi}{12}$ satisfies

$\therefore$ general solution is $x = 2n\pi + \frac{\pi}{12}, 2n\pi + \frac{17\pi}{12}$

15. Find the general solution of the trigonometric equation $3^{\left(\frac{1}{2} + \log_3(\cos x + \sin x)\right)} - 2^{\log_2(\cos x - \sin x)} = \sqrt{2}$.

त्रिकोणमितीय समीकरण $3^{\left(\frac{1}{2} + \log_3(\cos x + \sin x)\right)} - 2^{\log_2(\cos x - \sin x)} = \sqrt{2}$ का व्यापक हल ज्ञात कीजिए।

Ans. $x = 2n\pi + \frac{\pi}{12}$

Sol. $3^{\left(\frac{1}{2}+\log_3(\cos x+\sin x)\right)} - 2^{\log_2(\cos x-\sin x)} = \sqrt{2}$

$\because$ $\sin x + \cos x > 0$ & $\cos x - \sin x > 0$

$\Rightarrow \sin x + \cos x > 0$ & $\sin x - \cos x < 0$

In $(0, 2\pi)$

$$x \in \left(\frac{7\pi}{4}, \frac{\pi}{4}\right)$$

Now

$$\sqrt{3}(\sin x + \cos x) - (\cos x - \sin x) = \sqrt{2}$$

$$\Rightarrow \frac{\sqrt{3}-1}{2\sqrt{2}}\cos x + \frac{\sqrt{3}+1}{2\sqrt{2}}\sin x = \frac{1}{2}$$

$$\Rightarrow \cos\left(x - \frac{5\pi}{12}\right) = \frac{1}{2} = \cos\frac{\pi}{3}$$

$$\Rightarrow x = \left(2n\pi \pm \frac{\pi}{3}\right) + \frac{5\pi}{12}$$

$\because$ $\sin x + \cos x > 0$ & $\sin x - \cos x < 0$

$$\Rightarrow x = 2n\pi + \frac{\pi}{12}; n \in I$$

16. If sum of all the solutions of the equation $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}}\cos 2x$ is equal to $\frac{k\pi}{6}$, then find the value of k.

यदि समीकरण $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}}\cos 2x$ के सभी हलों का योग $\frac{k\pi}{6}$ है, तो k का मान ज्ञात कीजिये।

Ans. 18

Sol. $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}}\cos 2x$

For log

$\frac{9x-x^2-14}{7} > 0$ & $\frac{9x-x^2-14}{7} \neq 1$

$\Rightarrow x^2 - 9x + 14 < 0$ & $x^2 - 9x + 21 \neq 0$

$\Rightarrow (x-7)(x-2) < 0$ & $x^2 - 9x + 21 \neq 0$

$\Rightarrow x \in (2, 7)$

For log

$\sin 3x - \sin x > 0$ & $\cos 2x > 0$

$\Rightarrow 2\cos 2x \sin x > 0$ & $\cos 2x > 0$

$\Rightarrow \sin x > 0$ & $\cos 2x > 0$

Now

$\sin 3x - \sin x = \cos 2x$

$\Rightarrow 3\sin x - 4\sin^3 x - \sin x = 1 - 2\sin^2 x$

$\Rightarrow 4\sin^3 x - 2\sin^2 x - 2\sin x + 1 = 0$

$\Rightarrow (2\sin^2 x - 1)(2\sin x - 1) = 0$

$\Rightarrow \sin^2 x = \frac{1}{2}$ or $\sin x = \frac{1}{2}$

$\Rightarrow \sin^2 x = \sin^2 \frac{\pi}{4}$ or $\sin x = \frac{1}{2}$

$\Rightarrow x = n\pi \pm \frac{\pi}{4}; n \in I$ or $x = n\pi + (-1)^n \frac{\pi}{6}$

$\Rightarrow x = \frac{5\pi}{6}, \frac{13\pi}{6}$

$\Rightarrow \text{sum} = \frac{5\pi}{6} + \frac{13\pi}{6} = \frac{18\pi}{6}$

$\Rightarrow k = 18$

NUMBER OF SOLUTION IN AN INTERVAL

17. If the quadratic equation $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$ has two integral roots, then sum of all possible values of θ in interval $(0, 2\pi)$ is $k\pi$. Find the value of k.

यदि द्विघातीय समीकरण $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$ के दोनों मूल पूर्णांक है तो अन्तराल$(0, 2\pi)$ में θ के सारे सम्भावित मानों का योग $k\pi$ होगा तो k का मान ज्ञात करों।

Ans. 4

Sol. Given

Quadratic Equation

$x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$ has two integral roots.

Then $2 - \tan\theta \in Q$

& $1 + \tan\theta \in Q$

& $D \equiv (2 - \tan\theta)^2 + 4(1 + \tan\theta)$ must be perfect square.

$\Rightarrow D \equiv 4 + \tan^2\theta + 4$

$\Rightarrow D \equiv \tan^2\theta + 8$; is perfect square

if $\tan^2\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}, ...$

Given $\theta \in (0, 2\pi)$

$\Rightarrow$ Sum of all possible values of θ is

$\frac{\pi}{4} + \frac{5\pi}{4} + \frac{3\pi}{4} + \frac{7\pi}{4} = \frac{16\pi}{4} = 4\pi$

$\Rightarrow k = 4$

TRIGONOMETRIC INEQUALITIES

18. Solve the inequality $\sin 2x > \sqrt{2}\ \sin^2 x + (2 - \sqrt{2}\ )\cos^2 x$.

असमिका $\sin 2x > \sqrt{2}\ \sin^2 x + (2 - \sqrt{2}\ )\cos^2 x$ को हल करें।

Ans. $n\pi + \frac{\pi}{8} < x < n\pi + \frac{\pi}{4}$

Sol. $\sin 2x > \sqrt{2}\sin^2 x + \left(2 - \sqrt{2}\right)\cos^2 x$

$\Rightarrow 2\sin x \cos x > \sqrt{2}\sin^2 x + 2\cos^2 x - \sqrt{2}\cos^2 x$

$\Rightarrow \frac{2\sin x \cos x}{\cos^2 x} > \sqrt{2}\tan^2 x + 2 - \sqrt{2}$

$\Rightarrow \sqrt{2}\tan^2 x - 2\tan x + 2 - \sqrt{2} < 0$

$\sqrt{2}\left(\tan^2 x - 1\right) - 2\left(\tan x - 1\right) < 0$

$\Rightarrow \left(\tan x - 1\right)\left\{\sqrt{2}\left(\tan x + 1\right) - 2\right\} < 0$

$\Rightarrow \left(\tan x - 1\right)\left(\tan x + 1 - \sqrt{2}\right) < 0$

$\Rightarrow \tan x \in \left(\sqrt{2} - 1, 1\right)$

$\Rightarrow x \in \left(\frac{\pi}{8}, \frac{\pi}{4}\right)$

$\Rightarrow$ General solution is $x \in \left(n\pi + \frac{\pi}{8}, n\pi + \frac{\pi}{4}\right)$

SYSTEM OF TRIGONOMETRIC EQUATION

19. For $0 < x, y < \pi$, find the number of ordered pairs(x, y) satisfying the system of equations $\cot^2(x - y) - \left(1 + \sqrt{3}\right)\cot(x - y) + \sqrt{3} = 0$ and $\cos y = \frac{\sqrt{3}}{2}$.

यदि $0 < x, y < \pi$ हो, तो समीकरणों $\cot^2(x-y) - \left(1+\sqrt{3}\right)\cot(x-y) + \sqrt{3} = 0$ तथा $\cos y = \frac{\sqrt{3}}{2}$ के निकाय को संतुष्ट करने वाले क्रमित युग्मों (x, y) की संख्या ज्ञात कीजिये।

Ans. 2

Sol. Given $0 < x, y < \pi$

$$\cot^2(x-y) - \left(1+\sqrt{3}\right)\cot(x-y) + \sqrt{3} = 0$$

$$\& \ \cos y = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \cot^2(x-y) - \cot(x-y) - \sqrt{3}\cot(x-y) + \sqrt{3} = 0$$

$$\Rightarrow \left(\cot(x-y) - 1\right)\left(\cot(x-y) - \sqrt{3}\right) = 0$$

$\Rightarrow \cot(x-y) = 1$ or $\cot(x-y) = \sqrt{3}$

$\Rightarrow \tan(x-y) = 1$ or $\tan(x-y) = \frac{1}{\sqrt{3}}$

$\Rightarrow x - y = n\pi + \frac{\pi}{4}$ oρ $x - y = n\pi + \frac{\pi}{6}$

$\because \cos y = \frac{\sqrt{3}}{2} = \cos\left(\frac{\pi}{6}\right)$

$$\Rightarrow y = 2n\pi \pm \frac{\pi}{6}$$

$\because \ 0 < x, y < \pi$

$$\Rightarrow y = \frac{\pi}{6}, \frac{5\pi}{6}$$

If $y = \frac{\pi}{6}$ then

$\Rightarrow x = n\pi + \frac{\pi}{4} + \frac{\pi}{6}$ or $x = n\pi + \frac{\pi}{6} + \frac{\pi}{6}$

$\Rightarrow x = \frac{5\pi}{12}$ or $x = \frac{\pi}{3}$

If $y = \frac{5\pi}{6}$ then

$x = n\pi + \frac{\pi}{4} + \frac{5\pi}{6}$ or $x = n\pi + \pi$

$\because\ x \in (0, \pi)$

$\Rightarrow$ There is no x for $y = \frac{5\pi}{6}$

$\therefore$ Only possible pairs are $\left(\frac{5\pi}{12}, \frac{\pi}{6}\right), \left(\frac{\pi}{3}, \frac{\pi}{6}\right)$

$\Rightarrow$ No. of ordered pairs are 2.

20. If the number of solutions satisfying the equation $(\sin\theta - 1)(2\sin\theta - 1)(3\sin\theta - 1).....(n\sin\theta - 1) = 0$ (where $n \in N$) in $[0, \pi]$ is 9, then find the number of ordered pairs (x, y) satisfying the equation $3 + \text{cosec}^2 x + 2^{\sin^2 y} = n$, where $0 \le x, y \le 4\pi$.

यदि समीकरण $(\sin\theta - 1)(2\sin\theta - 1)(3\sin\theta - 1).....(n\sin\theta - 1) = 0$ (जहाँ $n \in N$) को संतुष्ट करने वाले हलों की संख्या अन्तराल $[0, \pi]$ में 9 है, तो समीकरण $3 + \text{cosec}^2 x + 2^{\sin^2 y} = n$, जहाँ $0 \le x, y \le 4\pi$, को संतुष्ट करने वाले क्रमित युग्मों (x, y) की संख्या ज्ञात कीजिए।

Ans. 20

Sol. Given numbers of solutions satisfying the equation $(\sin\theta - 1)(2\sin\theta - 1)$ ..... $(n\sin\theta - 1) = 0$ in $[0, \pi]$ is 9.

$\Rightarrow \sin\theta = 1$ or $\sin\theta = \frac{1}{2}$ or $\sin\theta = \frac{1}{3}$

$\downarrow$ $\downarrow$ $\downarrow$

one solution two solution two solution

or $\sin\theta = \frac{1}{4}$ or $\sin\theta = \frac{1}{5}$

$\downarrow$ $\downarrow$

two solution two solution

$\Rightarrow n = 5$

$3 + \text{cosec}^2 x + 2^{\sin^2 y} = n = 5; 0 \le x, y \le 4\pi$

$\Rightarrow \text{cosec}^2 x + 2^{\sin^2 y} = 2$

$\because\ \text{cosec}2x \ge 1$ & $2^{\sin^2 y} \ge 1$

$\therefore$ This is possible only when

$\text{cosec}^2 x = 1$ & $2^{\sin^2 y} = 1$

$\Rightarrow x = n\pi \pm \frac{\pi}{2}$ & $y = n\pi$

$\Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$ & $y = 0, \pi, 2\pi, 3\pi, 4\pi$

$\therefore$ Number of ordered pairs (x, y) is $4 \times 5 = 20$.

21. Solve x and y : $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$, $5.16^{\sin x} - 2.3^{\frac{1}{\cos y}} = 2$

x तथा y ज्ञात कीजिए : $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$, $5.16^{\sin x} - 2.3^{\frac{1}{\cos y}} = 2$

Ans. $x = n\pi + (-1)^n \frac{\pi}{6}$ and $y = 2m\pi \pm \frac{\pi}{3}$, where m, n $\in$ I

Sol. $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$

$5 \cdot 16^{\sin x} - 2 \cdot 3^{\frac{1}{\cos y}} = 2$

Let $4^{\sin x} = a$, $3^{\frac{1}{\cos y}} = b$

$\Rightarrow a + b = 11$ & $5 \cdot a^2 - 2b = 2$

$\Rightarrow 5a^2 - 2(11 - a) - 2 = 0$

$\Rightarrow 5a^2 + 2a - 24 = 0$

$\Rightarrow (5a + 12)(a - 2) = 0$

$\Rightarrow a = 2$ or $\underbrace{a = \frac{-12}{5}}_{\text{Not possible}}$

$\Rightarrow 4^{\sin x} = 2$ & $3^{\sec y} = 9 = 3^2$

$\Rightarrow \sin x = \frac{1}{2}$ & $\sec y = 2$

$\Rightarrow \sin x = \frac{1}{2}$ & $y = 2m\pi \pm \frac{\pi}{3}$

$\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}; n \in I$

& $y = 2m\pi \pm \frac{\pi}{3}; m \in I$

22. The number of values of θ in the interval $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for n = 0, ± 1, ± 2 and $\tan\theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$, is. **[JEE-2010]**

अन्तराल $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ में θ के मानों की संख्या जिसके लिये $\theta \neq \frac{n\pi}{5}$ जहाँ n = 0, ± 1, ± 2 तथा $\tan\theta = \cot 5\theta$ साथ ही $\sin 2\theta = \cos 4\theta$ है। **[JEE-2010]**

Ans. 3

Sol. Given $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ & $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm1, \pm2$

$\sin2\theta = \cos4\theta$

$\sin2\theta = 1 - 2\sin^2 2\theta$

$2\sin^2 2\theta + \sin2\theta - 1 = 0$

$\Rightarrow \sin 2\theta = -1$ or $\sin 2\theta = \frac{1}{2}$

$\therefore \theta = -\frac{\pi}{4}, \frac{\pi}{12}, \frac{5\pi}{12}$

Now $\tan\theta = \cot 5\theta$ (Given)

All three obtained values of θ satisy the given equation.

Hence, number of values of θ are 3.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 24

Let $f(x) = \sin^3 x + 8\cos^6 x - 3$, $g(x) = 2 + 3\sin2x\cos x$ and $h(x) = \sin x + 3\cos2x\sin x$.
On the basis of above information, answer the following questions :

माना $f(x) = \sin^3 x + 8\cos^6 x - 3$, $g(x) = 2 + 3\sin2x\cos x$ तथा $h(x) = \sin x + 3\cos2x\sin x$.
उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उतर दीजिए।

23. If $g(x) + h(x) = 0$, then number of solutions in the interval $x \in [0,2\pi]$ is :

यदि $g(x) + h(x) = 0$, तो अन्तराल $x \in [0,2\pi]$ में हलों की संख्या कितनी होगी :

(A) 1 (B) 3 (C) 5 (D) 6

Ans. C

Sol. $g(x) + h(x) = 0$

$\Rightarrow 2 + 3(\sin 2x \cos x + \cos 2x \sin x) + \sin x = 0$

$\Rightarrow 2 + 3\sin 3x + \sin x = 0$

$\Rightarrow 9\sin x - 12\sin^3 x + \sin x + 2 = 0$

$\Rightarrow 12\sin^3 x - 10\sin x - 2 = 0$

$\Rightarrow 6\sin^3 x - 5\sin x - 1 = 0$

$\Rightarrow (\sin x - 1)(6\sin^2 x + 6\sin x + 1) = 0$

$\Rightarrow \sin x = 1$ or $\sin x = \frac{-3-\sqrt{3}}{6}$ or $\sin x = \frac{-3+\sqrt{3}}{6}$

In $0 \le x \le 2\pi$

$\sin x = 1$ $\rightarrow$ only one solution

$\sin x = \frac{-3-\sqrt{3}}{6}$ $\rightarrow$ two solutions

$\sin x = \frac{-3+\sqrt{3}}{6}$ $\rightarrow$ two solutions

$\therefore$ No. of solution is 5.

24. If $f(x)+g(x)=0$, then number of solutions in the interval $x \in \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$ is :

यदि $f(x)+g(x)=0$, तो अन्तराल $x \in \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$ में हलों की संख्या कितनी होगी :

(A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. $f(x)+g(x)=0$; $x \in \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$

$\Rightarrow \sin^3 x + 8\cos^6 x - 3 + 2 + 3\sin 2x \cos x = 0$

$\Rightarrow (\sin x)^3 + (2\cos^2 x)^3 + (-1)^3 = 3\sin x (2\cos^2 x)(-1)$

$\Rightarrow \sin x + 2\cos^2 x - 1 = 0$ or $\underbrace{\sin x = -1 = 2\cos^2 x}_{\text{Not possible}}$ $\left\{\because a^3+b^3+c^3 = 3abc \text{ if } a=b=c \text{ or } a+b+c=0\right\}$

$\Rightarrow \sin x + 2(1-\sin^2 x) - 1 = 0$

$\Rightarrow 2\sin^2 x - \sin x - 1 = 0$

$\Rightarrow (\sin x - 1)(2\sin x + 1) = 0$

$\Rightarrow \sin x = 1$ or $\sin x = \frac{-1}{2}$

In $\left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$

Number of solutions is 5.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

25. If $\left(\sin 2x + \sqrt{3}\cos 2x\right)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ then the value of cos x can be

यदि $\left(\sin 2x + \sqrt{3}\cos 2x\right)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ हो, तो cos x का मान हो सकता है –

(A) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (B) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (C) $-\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)$ (D) $\frac{\sqrt{2}-\sqrt{6}}{4}$

Ans. BD

Sol. $\left(\sin 2x+\sqrt{3}\cos 2x\right)^2=5+\cos\left(2x-\frac{\pi}{6}\right)$

$\Rightarrow 4\left(\frac{\sin 2x}{2}+\frac{\sqrt{3}}{2}\cos 2x\right)^2=5+\cos\left(2x-\frac{\pi}{6}\right)$

$\Rightarrow 4\cos^2\left(2x-\frac{\pi}{6}\right)-\cos\left(2x-\frac{\pi}{6}\right)=5$

$\Rightarrow 4\cos^2\left(2x-\frac{\pi}{6}\right)-\cos\left(2x-\frac{\pi}{6}\right)-5=0$

$\Rightarrow \left(4\cos\left(2x-\frac{\pi}{6}\right)-5\right)\left(\cos\left(2x-\frac{\pi}{6}\right)+1\right)=0$

$\Rightarrow \underbrace{\cos\left(2x-\frac{\pi}{6}\right)=\frac{5}{4}}_{\text{Not possible as } \frac{5}{4}>1.}$ or $\cos\left(2x-\frac{\pi}{6}\right)=-1$

$\Rightarrow \cos\left(2x-\frac{\pi}{6}\right)=-1$

$\Rightarrow 2x-\frac{\pi}{6}=(2n-1)\pi$

$\Rightarrow x=(2n-1)\frac{\pi}{2}+\frac{\pi}{12}$

$\Rightarrow x=n\pi-\frac{5\pi}{12};n\in I$

$\Rightarrow x=-\frac{5\pi}{12},\frac{7\pi}{12},\frac{-17\pi}{12},.....$

$\cos\left(\frac{7\pi}{12}\right)=\cos\left(\pi-\frac{5\pi}{12}\right)=-\cos\frac{5\pi}{12}=\frac{\sqrt{2}-\sqrt{6}}{4}$

$\cos\left(-\pi-\frac{5\pi}{12}\right)=\cos\left(\pi+\frac{5\pi}{12}\right)=-\cos\left(\frac{5\pi}{12}\right)=\frac{\sqrt{6}-\sqrt{2}}{4}$

26. If $|\cos x|^{\sin^2 x-\frac{3}{2}\sin x+\frac{1}{2}}=1$. Then possible values of x are :

यदि $|\cos x|^{\sin^2 x-\frac{3}{2}\sin x+\frac{1}{2}}=1$ तब x के सम्भव मान होंगे :

(A) $n\pi$ or $n\pi + (-1)^n \frac{\pi}{6}, n \in I$ (B) $n\pi$ or $2n\pi + \frac{\pi}{2}$ or $n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(C) $n\pi + (-1)^n \frac{\pi}{6}, n \in I$ (D) $n\pi, n \in I$

Ans. ACD

Sol. $|\cos x|^{\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}} = 1$

Case I :- $\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2} = 0$

$\Rightarrow 2\sin^2 x - 3\sin x + 1 = 0$

$\Rightarrow (2\sin x - 1)(\sin x - 1) = 0$

$\Rightarrow \sin x = \frac{1}{2}$ or $\sin x = 1$

$\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}$ or $x = (4n+1)\frac{\pi}{2}; n \in I$

Case II :- $|\cos x| = 1$

$\Rightarrow x = n\pi$

27. If $\frac{\sin^2 2x + 4\sin^4 x - 4\sin^2 x \cos^2 x}{4 - \sin^2 2x - 4\sin^2 x} = \frac{1}{9}$ and $0 < x < \pi$, then value of x is :

यदि $\frac{\sin^2 2x + 4\sin^4 x - 4\sin^2 x \cos^2 x}{4 - \sin^2 2x - 4\sin^2 x} = \frac{1}{9}$ तथा $0 < x < \pi$, तब x के सम्भव मान होंगे :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

Ans. BD

Sol. $\frac{\sin^2 2x + 4\sin^4 x - 4\sin^2 x \cos^2 x}{4 - \sin^2 2x - 4\sin^2 x} = \frac{1}{9}; 0 < x < \pi$

$\Rightarrow \frac{\sin^2 2x + 4\sin^4 x - \sin^2 2x}{4\left(1 - \sin^2 x \cos^2 x - \sin^2 x\right)} = \frac{1}{9}$

$\Rightarrow 9\sin^4 x = 1 - \sin^2 x + \sin^4 x - \sin^2 x$

$\Rightarrow 8\sin^4 x = 1 - 2\sin^2 x$

$\Rightarrow 8\sin^4 x + 2\sin^2 x - 1 = 0$

$\Rightarrow 8\sin^4 x + 4\sin^2 x - 2\sin^2 x - 1 = 0$

$\Rightarrow (4\sin^2 x - 1)(2\sin^2 x + 1) = 0$

$\Rightarrow \sin^2 x = \frac{1}{4}$ or $\underbrace{\sin^2 x = \frac{-1}{2}}_{\text{Not possible}}$

$\Rightarrow x = n\pi \pm \frac{\pi}{6}$

$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$ $\{\because 0 < x < \pi\}$

28. $5 \sin x - 12 \cos x = -13 \sin 3x$, if $\phi = \sin^{-1}\left(\frac{12}{13}\right)$, then which of the following options are correct :

यदि $\phi = \sin^{-1}\left(\frac{12}{13}\right)$, तथा $5 \sin x - 12 \cos x = -13 \sin 3x$ हो तो निम्न में से कौनसा विकल्प सही है :

(A) $\sin\left(2x - \frac{\phi}{2}\right) = 0$ (B) $\sin\left(x + \frac{\phi}{2}\right) = 0$ (C) $\cos\left(x + \frac{\phi}{2}\right) = 0$ (D) $\cos\left(2x + \frac{\phi}{2}\right) = 0$

Ans. AC

Sol. $5 \sin x - 12 \cos x = -13 \sin 3x$ & $\phi = \sin^{-1}\left(\frac{12}{13}\right)$

$\Rightarrow \frac{5}{13}\sin x - \frac{12}{13}\cos x = -\sin 3x$

$\Rightarrow \sin x \cos\phi - \cos x \sin\phi = -\sin 3x$

$\Rightarrow \sin(x - \phi) + \sin 3x = 0$

$\Rightarrow 2\sin\left(\frac{4x - \phi}{2}\right)\cos\left(\frac{2x + \phi}{2}\right) = 0$

$\Rightarrow \sin\left(2x - \frac{\phi}{2}\right) = 0$ or $\cos\left(x + \frac{\phi}{2}\right) = 0$

29. If $\sin 5\theta = a \sin^5\theta + b \sin^3\theta + c \sin\theta + d$, $\forall\ \theta \in R$, then :

यदि $\sin 5\theta = a \sin^5\theta + b \sin^3\theta + c \sin\theta + d$, $\forall\ \theta \in R$ तब :

(A) $a + b + c + d = 1$ (B) $a + b + c = 1$ (C) $5a + 4b = 0$ (D) $b + 4c = 0$

Ans. ABCD

Sol. Given $\sin 5\theta = a \sin^5\theta + b \sin^3\theta + c \sin\theta + d$ $\forall\theta \in R$

$\Rightarrow \sin 5\theta = \sin(3\theta + 2\theta)$

$= \sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$

$= 2\sin\theta \cos\theta (4\cos^3\theta - 3\cos\theta) + (1 - 2\sin^2\theta)(3\sin\theta - 4\sin^3\theta)$

$= 8 \sin\theta \cos^4\theta - 6 \sin\theta \cos^2\theta + 3 \sin\theta - 4 \sin^3\theta - 6 \sin^3\theta + 8 \sin^5\theta$

$= 8\sin\theta + 8\sin^5\theta - 16\sin^3\theta - 3\sin\theta - 4\sin^3\theta + 8\sin^5\theta$

$= 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$

$\Rightarrow a = 16, b = -20, c = 5, d = 0$

$\Rightarrow a + b + c + d = 1$

$a + b + c = 1$

$5a + 4b = 80 - 80 = 0$

& $b + 4c = 0$

MATCH THE COLUMN TYPE QUESTION(S)

30. Match the following column :

Column I	**Column II**
(A) Solution of $\lvert\tan\theta\rvert + \lvert\cot\theta\rvert \le 2$ is	(P) $2n\pi + \frac{5\pi}{4}$, $n \in I$
(B) Solution of $\left(\sqrt{2}\cos\theta + 1\right)^2 + (\tan\theta - 1)^2 = 0$ is	(Q) Null set
(C) Solution of $\sin\left(\frac{11}{10}\cos\theta\right) = \cos\left(\frac{11}{10}\sin\theta\right)$ is	(R) $(2n+1)\frac{\pi}{4}$, $n \in I$
(D) Solution of $\sin x = 2^{x-1} + 2^{1-x}$ is	(S) $n\pi + (-1)^n\frac{\pi}{6}$, $n \in I$

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) $\lvert\tan\theta\rvert + \lvert\cot\theta\rvert \le 2$ का हल होगा	(P) $2n\pi + \frac{5\pi}{4}$, $n \in I$
(B) $\left(\sqrt{2}\cos\theta + 1\right)^2 + (\tan\theta - 1)^2 = 0$ का हल होगा	(Q) रिक्त समुच्चय
(C) $\sin\left(\frac{11}{10}\cos\theta\right) = \cos\left(\frac{11}{10}\sin\theta\right)$ का हल होगा	(R) $(2n+1)\frac{\pi}{4}$, $n \in I$
(D) $\sin x = 2^{x-1} + 2^{1-x}$ का हल होगा	(S) $n\pi + (-1)^n\frac{\pi}{6}$, $n \in I$

Ans. (A) R, (B) P, (C) Q, (D) Q

Sol. (A) $|\tan\theta| + |\cot\theta| \le 2$

Let $\tan\theta = t$

$\Rightarrow |t| + \frac{1}{|t|} \le 2$ $\quad\left\{\because t + \frac{1}{t} \ge 2 \;\&\; t + \frac{1}{t} \le -2\right.$

$\Rightarrow |t| + \frac{1}{|t|} = 2$

$\Rightarrow (|t| - 1)^2 = 0$

$\Rightarrow \tan\theta = 1$ or $\tan\theta = -1$

$\Rightarrow \theta = n\pi + \frac{\pi}{4}$ or $\theta = n\pi - \frac{\pi}{4}$

For n = 0

$\theta = \frac{\pi}{4}$ or $\theta = -\frac{\pi}{4}$

For n = 1

$\theta = \frac{5\pi}{4}$ or $\theta = \frac{3\pi}{4}$

$\Rightarrow \theta = (2n+1)\frac{\pi}{4}; n \in I$

(B) $\left(\sqrt{2}\cos\theta + 1\right)^2 + (\tan\theta - 1)^2 = 0$

$\Rightarrow \tan\theta = 1$ & $\sqrt{2}\cos\theta + 1 = 0$

$\Rightarrow \tan\theta = 1$ & $\cos\theta = \frac{-1}{\sqrt{2}}$

$\Rightarrow \theta = n\pi + \frac{\pi}{4}$ & $\theta = 2n\pi - \frac{3\pi}{4}; n \in I$

$\Rightarrow \theta = 2n\pi + \frac{5\pi}{4}; n \in I$

(C) $\sin\left(\frac{11}{10}\cos\theta\right) = \cos\left(\frac{11}{10}\sin\theta\right)$

$\Rightarrow \frac{11}{10}\cos\theta = \frac{\pi}{2} - \frac{11}{10}\sin\theta$

$\Rightarrow \frac{11}{10}(\cos\theta + \sin\theta) = \frac{22}{14}$

$\Rightarrow \cos\theta + \sin\theta = \frac{10}{7}$

$\because \cos\theta + \sin\theta \in \left[-\sqrt{2}, \sqrt{2}\right]$

but $\frac{10}{7} > \sqrt{2}$

$\therefore$ There is no solution.

(D) $\sin x = 2^{x-1} + 2^{1-x}$

$$\Rightarrow \underbrace{\sin x}_{\substack{\downarrow \\ [-1,1]}} = \underbrace{\frac{2^x}{2} + \frac{2}{2^x}}_{\in[2,\infty)}$$

$\Rightarrow$ No solution

31. Match the following column :

Column I (Equation)	**Column II (General Solution)**
(A) $\sin x + \cos x = 1$	(P) $2n\pi, n \in I$
(B) $\sin x + \cos x = -1$	(Q) $(2n+1)\pi, n \in I$
(C) $\lvert\sin x\rvert + \lvert\cos x\rvert = 1$	(R) $(4n+1)\frac{\pi}{2}, n \in I$
(D) $\lvert\sin x + \cos x\rvert = 1$	(S) $(4n-1)\frac{\pi}{2}, n \in I$

निम्न कॉलम को सुमेलित कीजिए :

Column I (समीकरण)	**Column II (व्यापक हल)**
(A) $\sin x + \cos x = 1$	(P) $2n\pi, n \in I$
(B) $\sin x + \cos x = -1$	(Q) $(2n+1)\pi, n \in I$
(C) $\lvert\sin x\rvert + \lvert\cos x\rvert = 1$	(R) $(4n+1)\frac{\pi}{2}, n \in I$
(D) $\lvert\sin x + \cos x\rvert = 1$	(S) $(4n-1)\frac{\pi}{2}, n \in I$

Ans. (A) PR, (B) QS, (C) PQRS, (D) PQRS

Sol. (A) $\sin x + \cos x = 1$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \cos\frac{\pi}{4}$$

$$\Rightarrow x = \left(2n\pi \pm \frac{\pi}{4}\right) + \frac{\pi}{4}; n \in I$$

$\Rightarrow x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{2}; n \in I$

(B) $\sin x + \cos x = -1$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \cos\frac{3\pi}{4}$$

$$\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}$$

$\Rightarrow x = (2n+1)\pi$ or $x = (4n-1)\frac{\pi}{2}; n \in I$

(C) $|\sin x| + |\cos x| = 1$

In first quad.

$\sin x + \cos x = 1$

$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \cos\frac{\pi}{4}$

$\Rightarrow x = 2n\pi$ or $(4n+1)\frac{\pi}{2}; n \in I$

In second quad.

$\sin x - \cos x = 1$

$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = \cos\frac{3\pi}{4}$

$\Rightarrow x = (4n+1)\frac{\pi}{2}$ or $x = (2n-1)\pi; n \in I$

In third quad.

$\sin x + \cos x = -1$

$x = (2n+1)\pi$ or $x = (4n-1)\frac{\pi}{2}; n \in I$

In fourth quad.

$-\sin x + \cos x = 1$

$\Rightarrow \cos x - \sin x = 1$

$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = \cos\frac{\pi}{4}$

$\Rightarrow x = 2n\pi; n \in I$ or $x = (4n-1)\frac{\pi}{2}; n \in I$

(D) $|\sin x + \cos x| = 1$

$\Rightarrow \sin x + \cos x = 1$ or $\sin x + \cos x = -1$

$\Rightarrow x = 2n\pi$ or $x = (2n+1)\pi$ or $x = (4n+1)\frac{\pi}{2}$ or $x = (4n-1)\frac{\pi}{2}$

32. If α and β are the roots of the equation, $a\cos\theta + b\sin\theta = c$ then match the entries of column-I with the entries of column-II :

यदि α तथा β समीकरण $a\cos\theta + b\sin\theta = c$ के मूल है तो स्तम्भ का मिलान कीजिए :

Column-I	**Column-II**
(A) $\sin\alpha + \sin\beta$	(P) $\frac{2b}{a+c}$
(B) $\sin\alpha \,.\, \sin\beta$	(Q) $\frac{c-a}{c+a}$
(C) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2}$	(R) $\frac{2bc}{a^2+b^2}$
(D) $\tan\frac{\alpha}{2} \,.\, \tan\frac{\beta}{2} =$	(S) $\frac{c^2-a^2}{a^2+b^2}$

Ans. (A) R; (B) S; (C) P; (D) Q

Sol. $a\cos\theta + b\sin\theta = c$

$\Rightarrow a\cos\theta = c - b\sin\theta$

square both sides

$\Rightarrow a^2\cos^2\theta = c^2 + b^2\sin^2\theta - 2bc\sin\theta$

$$\Rightarrow \underbrace{\left(a^2+b^2\right)\sin^2\theta - 2bc\sin\theta + \left(c^2-a^2\right) = 0}_{\text{Quadratic Equation have roots } \sin\alpha \text{ \& } \sin\beta}$$

$$\Rightarrow \sin\alpha + \sin\beta = \frac{2bc}{a^2+b^2}$$

$$\& \ \sin\alpha\cdot\sin\beta = \frac{c^2-a^2}{a^2+b^2}$$

Now $a\cos\theta + b\sin\theta = c$

$$\Rightarrow \frac{a\left(1-\tan^2\frac{\theta}{2}\right)}{1+\tan^2\frac{\theta}{2}} + \frac{2b\left(\tan\frac{\theta}{2}\right)}{1+\tan^2\frac{\theta}{2}} = c$$

Let $\tan\frac{\theta}{2} = t$

$$\Rightarrow \frac{a\left(1-t^2\right)}{\left(1+t^2\right)} + \frac{2bt}{\left(1+t^2\right)} = c$$

$\Rightarrow a - at^2 + 2bt - c - ct^2 = 0$

$\Rightarrow (a+c)t^2 - 2bt + (c-a) = 0$

$$\Rightarrow \underbrace{(a+c)\tan^2\frac{\theta}{2} - 2b\tan\frac{\theta}{2} + c - a = 0}_{\text{Quadratic Equation have roots } \tan\frac{\alpha}{2} \text{ \& } \tan\frac{\beta}{2}}$$

$$\Rightarrow \tan\frac{\alpha}{2} + \tan\frac{\beta}{2} = \frac{2b}{a+c}$$

$$\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2} = \frac{c-a}{a+c}$$

Exercise J-M

1. The number of solutions of $\sin^7 x + \cos^7 x = 1, x \in [0, 4\pi]$ is equal to : **[JEE(Main)July2021]**

समीकरण, $\sin^7 x + \cos^7 x = 1$ के $x \in [0, 4\pi]$ में हलों की संख्या है : **[JEE(Main)July2021]**

(A) 5 (B) 7 (C) 9 (D) 11

Ans. A

Sol. $\sin^7 x + \cos^7 x = 1 : x \in [0, 4\pi]$

Since

$\sin^2 x \le 1$ & $\cos^2 x \le 1$

Also

$\sin^7 x \le \sin^2 x$

$\cos^7 x \le \cos^2 x$

by adding

$\sin^7 x + \cos^7 x \le 1$ ....(1)

but $\sin^7 x + \cos^7 x = 1$ ....(2)

hence by (1) and (2)

$\sin^7 x = \sin^2 x$ ....(3)

$\cos^7 x = \cos^2 x$ ....(4)

then all solution of (3) & (4) are

$x = 0,\ 2\pi,\ 4\pi,\ \frac{\pi}{2},\ \frac{5\pi}{2}$

$\therefore$ No. of solutions = 5

2. The sum of all values of x in $[0, 2\pi]$, for which $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$, is equal to : **[JEE(Main)July2021]**

$[0, 2\pi]$ में x के सभी मानों, जिनके लिए $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$ है, का योग हैं : **[JEE(Main)July2021]**

(A) 9π (B) 11π (C) 12π (D) 8π

Ans. A

Sol. $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$

$$\Rightarrow \frac{\sin 2x.\sin\left(\frac{5x}{2}\right)}{\sin\left(\frac{x}{2}\right)} = 0$$

Case-I

If $\sin 2x = 0$

Then $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$

Case-II

If $\sin\frac{5x}{2}=0$

Then $x=0,\frac{2\pi}{5},\frac{4\pi}{5},\frac{6\pi}{5},\frac{8\pi}{5},2\pi$

$\therefore$ solution of $\sin\frac{x}{2}=0$ are valid, because $x=0$, & 2π satisfy the given equation.

So solutions are $x=0,\frac{2\pi}{5},\frac{4\pi}{5},\frac{6\pi}{5},\frac{8\pi}{5},2\pi,\frac{\pi}{2},\pi,\frac{3\pi}{2}$

$\therefore$ Sum of solutions is 9π.

3. The sum of solutions of the equations $\frac{\cos x}{1+\sin x}=|\tan 2x|,\ x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)-\left\{\frac{\pi}{4},-\frac{\pi}{4}\right\}$ is :

[JEE(Main)Aug.2021]

समीकरण $\frac{\cos x}{1+\sin x}=|\tan 2x|,\ x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)-\left\{\frac{\pi}{4},-\frac{\pi}{4}\right\}$ के हलों का योग है : **[JEE(Main)Aug.2021]**

(A) $-\frac{11\pi}{30}$ (B) $-\frac{7\pi}{30}$ (C) $-\frac{\pi}{15}$ (D) $\frac{\pi}{10}$

Ans. A

Sol. $\frac{\cos x}{1+\sin x}=|\tan 2x|$

$\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)=|\tan 2x|$

C–1 If $\tan 2x\geq 0$

then $\tan 2x=\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)$

$\Rightarrow 2x=n\pi+\frac{\pi}{4}-\frac{x}{2}$

$\Rightarrow \frac{5x}{2}=n\pi+\frac{\pi}{4}$

$\Rightarrow x=\frac{2n\pi}{5}+\frac{\pi}{10}=\frac{(4n+1)\pi}{10}$

$\because \tan 2x\geq 0$

$\Rightarrow x = \frac{\pi}{10}$, $-\frac{3\pi}{10}$

C–2 If $\tan 2x < 0$ then $\tan 2x = -\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$

$\Rightarrow \tan 2x = \tan\left(\frac{x}{2} - \frac{\pi}{4}\right)$

$\Rightarrow 2x = n\pi + \frac{x}{2} - \frac{\pi}{4}$

$\Rightarrow \frac{3x}{2} = n\pi - \frac{\pi}{4} = \frac{(4n-1)\pi}{4}$

$\Rightarrow x = \frac{(4n-1)\pi}{6}$

$\Rightarrow x = \frac{-\pi}{6}$

$\therefore$ Sum of solution $= \frac{\pi}{10} - \frac{3\pi}{10} - \frac{\pi}{6} = -\frac{11\pi}{30}$

4. The number of solutions of the equation $32^{\tan^2 x} + 32^{\sec^2 x} = 81, 0 \le x \le \frac{\pi}{4}$ is : **[JEE(Main)Aug.2021]**

समीकरण $32^{\tan^2 x} + 32^{\sec^2 x} = 81, 0 \le x \le \frac{\pi}{4}$ के हलों की संख्या है : **[JEE(Main)Aug.2021]**

(A) 3 (B) 2 (C) 0 (D) 1

Ans. D

Sol. $(32)^{\tan^2 x} + (32)^{\sec^2 x} = 81$

$\Rightarrow (32)^{\tan^2 x} + 32 \times 32^{\tan^2 x} = 81$

$\Rightarrow 33 \times (32)^{\tan^2 x} = 81$

$\Rightarrow (32)^{\tan^2 x} = \frac{81}{33} = \frac{27}{11}$

$\Rightarrow (32)^{\tan^2 x} = \frac{27}{11}$

$\because x \in \left[0, \frac{\pi}{4}\right]$

$\Rightarrow \tan^2 x \in [0,1]$

$\Rightarrow 32^{\tan^2 x} \in [1,32]$

$\therefore$ one solution

5. If n is the number of solutions of the equation $2\cos x\left(4\sin\left(\frac{\pi}{4}+x\right)\sin\left(\frac{\pi}{4}-x\right)-1\right)=1$, $x \in [0,\pi]$ and S is the sum of all these solutions, then the ordered pair (n, S) is : **[JEE(Main)Aug.2021]**

यदि समीकरण $2\cos x\left(4\sin\left(\frac{\pi}{4}+x\right)\sin\left(\frac{\pi}{4}-x\right)-1\right)=1$, $x \in [0,\pi]$ के हलों की संख्या n है तथा S इन सभी हलों का योगफल है, तब क्रमित युग्म (n, S) है : **[JEE(Main)Aug.2021]**

(A) $(2, 8\pi/9)$ (B) $(3, 5\pi/3)$ (C) $(3, 13\pi/9)$ (D) $(2, 2\pi/3)$

Ans. C

Sol. $2\cos x\left(4\left(\sin^2\frac{\pi}{4}-\sin^2 x\right)-1\right)=1$

$\Rightarrow 2\cos x\left(4\left(\frac{1}{2}-\sin^2 x\right)-1\right)=1$

$\Rightarrow 2\cos x\ (2-4\sin^2 x-1)=1$

$\Rightarrow 2\cos x\ (1-4\ (1-\cos^2 x)=1$

$\Rightarrow 2\cos x\ (4\cos^2 x-3)=1$

$\Rightarrow (4\cos^3 x-3\cos x)=\frac{1}{2}$

$\Rightarrow \cos 3x=\frac{1}{2}$ .....(1)

$\because\ x \in [0,\pi]$

$\Rightarrow x=\frac{\pi}{9},\frac{5\pi}{9},\frac{7\pi}{9}$

Sum of all there solutions is $\frac{13\pi}{9}$

$\Rightarrow (n, S)=\left(3,\frac{13\pi}{9}\right)$

6. Let S be the sum of all solutions (in radians) of the equation $\sin^4\theta+\cos^4\theta-\sin\theta\cos\theta=0$ in $[0, 4\pi]$. Then $\frac{8S}{\pi}$ is equal to _____. **[JEE(Main)Aug.2021]**

माना $[0, 4\pi]$ में समीकरण $\sin^4\theta + \cos^4\theta - \sin\theta\cos\theta = 0$ के सभी हलों (रेडियन में) का योग S है। तो $\frac{8S}{\pi}$ बराबर है_____

[JEE(Main)Aug.2021]

Ans. 56

Sol. $\sin^4\theta + \cos^4\theta - \sin\theta\cos\theta = 0$

$\Rightarrow 1 - 2\sin^2\theta\cos^2\theta - \sin\theta\cos\theta = 0$

$\Rightarrow 1 - \frac{1}{2}\sin^2 2\theta - \frac{1}{2}\sin 2\theta = 0$

Let $\sin 2\theta = y$

$\Rightarrow y^2 + y - 2 = 0$

If $y = -2$ then $\underbrace{\sin 2\theta = -2}_{\text{Not possible}}$

If $y = 1$ then $\sin 2\theta = 1$

$\Rightarrow 2\theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}$

$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$

$\therefore \text{Sum} = \frac{\pi + 5\pi + 9\pi + 13\pi}{4} = 7\pi = S$

$\Rightarrow \frac{8S}{\pi} = 56$

7. Let $S = \left\{\theta, \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$. If $T = \sum_{\theta\in S}\cos 2\theta$, then T + n (S) is equal to :

[JEE(Main)June2022]

माना $S = \left\{\theta, \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$ है। यदि $T = \sum_{\theta\in S}\cos 2\theta$ है, तो T + n (S) बराबर है :

[JEE(Main)June2022]

(1) $7 + \sqrt{3}$ (2) 9 (3) $8 + \sqrt{3}$ (4) 10

Ans. Official Answer NTA (2)

Sol. $\sin\theta\tan\theta + \tan\theta = \sin 2\theta$

$s\left(\frac{s}{c}\right) + \frac{s}{c} = 2sc$

$\sin\theta = 0$ $\frac{s+1}{c} = 2c$

$\theta = -\pi, 0, \pi$

$s + 1 = 2c^2$

$s + 1 = 2 - 2s^2$

$2s^2 + s - 1 = 0$

$(s + 1)(2s - 1) = 0$

$\sin\theta = -1$ $\qquad$ $\sin\theta = \frac{1}{2}$

(Rejected) $\qquad$ $\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$S = \left\{-\pi, 0, \pi, \frac{\pi}{6}, \frac{5\pi}{6}\right\}$

$n(S) = 5$

$T = \cos(-2\pi) + \cos 0 + \cos 2\pi + \cos\frac{\pi}{3} + \cos\frac{5\pi}{3}$

$= 4$

$T + n(S) = 9$

8. The number of solutions of the equation $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is :

[JEE(Main)June2022]

समीकरण $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ के हलों की संख्या है :

[JEE(Main)June2022]

(1) 8 $\qquad$ (2) 5 $\qquad$ (3) 6 $\qquad$ (4) 7

Ans. Official Answer NTA (4)

Sol. $\cos\left(\frac{\pi}{3} + x\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$

$\cos^2\frac{\pi}{3} - \sin^2 x = \frac{1}{4}\cos^2 2x$

$\frac{1}{4} - \sin^2 x = \frac{1}{4}(1 - 2\sin^2 x)^2$

$(1 - 4\sin^2 x) = (1 - 2\sin^2 x)^2$

$\sin^2 x = t$

$(1 - 4t) = (1 - 2t)^2$

$1 - 4t = 4t^2 - 4t + 1$

$t^2 = 0$

$\sin^2 x = 0$

$x = -3\pi, -2\pi, -\pi, 0, \pi, 2\pi, 3\pi$

hence total number of solution are 7

9. The number of values of x in the interval $\left(\frac{\pi}{4},\frac{7\pi}{4}\right)$ for which $14\ \text{cosec}^2x - 2\sin^2x = 21 - 4\cos^2x$ holds, is _____.

[JEE(Main)June2022]

अंतराल $\left(\frac{\pi}{4},\frac{7\pi}{4}\right)$ में x के मानों, जिनके लिए $14\ \text{cosec}^2x - 2\sin^2x = 21 - 4\cos^2x$ है, की संख्या है_____

[JEE(Main)June2022]

Ans. Official Answer NTA (4)

Sol. $x \in \left(\frac{\pi}{4},\frac{7\pi}{4}\right)$

$14\,\text{cosec}^2 x - 2\sin^2 x = 21 - 4\cos^2 x$

$= 21 - 4\left(1 - \sin^2 x\right)$

$= 17 + 4\sin^2 x$

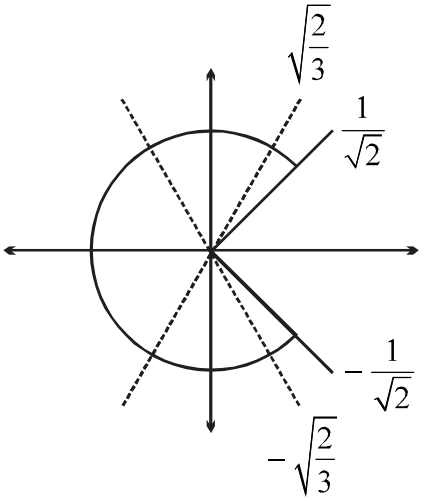
$14\,\text{cosec}^2 x - 6\sin^2 x = 17$

let $\sin^2 x = p \Rightarrow \frac{14}{p} - 6p = 17 \Rightarrow 14 - 6p^2 = 17p$

$6p^2 + 17p - 14 = 0$

$p = -3.5, \frac{2}{3} \Rightarrow \sin^2 x = \frac{2}{3}$

$\Rightarrow \sin x = \pm\sqrt{\frac{2}{3}}$



$\therefore$ Total 4 solutions

10. The number of elements in the set $S = \{\theta \in [-4\pi, 4\pi] : 3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0\}$ is _____.

[JEE(Main)June2022]

समुच्चय $S = \{\theta \in [-4\pi, 4\pi] : 3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0\}$ में अवयवों की संख्या _____

[JEE(Main)June2022]

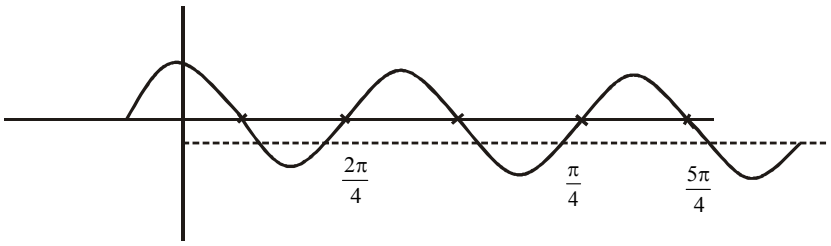
Ans. Official Answer NTA (32)

Sol. $S = \left\{\theta \in [-4\pi, 4\pi] : 3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0\right\}$

$3\cos^2 2\theta + 6\cos 2\theta - 5\left(1 + \cos 2\theta\right) + 5 = 0$

$\cos 2\theta\left(3\cos 2\theta + 1\right) = 0$

16 sol. $\cos 2\theta = 0$ $\quad \theta \in [-4\pi, 4\pi]$, $\cos 2\theta = \frac{-1}{3}$ 16 sol.



Total No. of elements in S = 32

11. The number of solutions of the equation $2\theta - \cos^2\theta + \sqrt{2} = 0$ in R is equal to _____.

[JEE(Main)June2022]

R में $2\theta - \cos^2\theta + \sqrt{2} = 0$ के हलों की संख्या है _____ **[JEE(Main)June2022]**

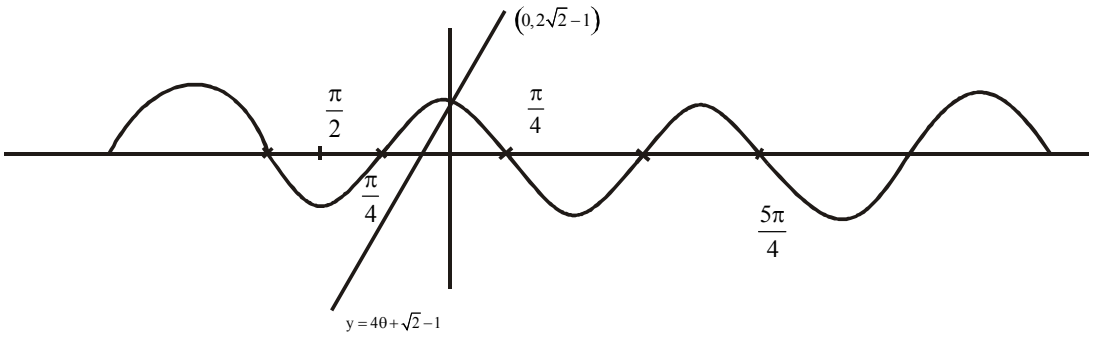
Ans. Official Answer NTA (1)

Sol. $2\theta - \cos^2\theta + \sqrt{2} = 0$

$$2\theta + \sqrt{2} = \frac{1+\cos 2\theta}{2}$$

$$4\theta + 2\sqrt{2} - 1 = \cos 2\theta$$

No. of sol. = No. of intersection parts of $y = \cos 2\theta$ and $y = 4\theta + 2\sqrt{2} - 1$



No. of sol. = 1

12. The number of solutions of the equation $\sin x = \cos^2 x$ in the interval (0, 10) is _____.

[JEE(Main)June2022]

अंतराल (0, 10) में समीकरण $\sin x = \cos^2 x$ के हलों की संख्या है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (4)

Sol. $\Rightarrow \sin x = \cos^2 x$

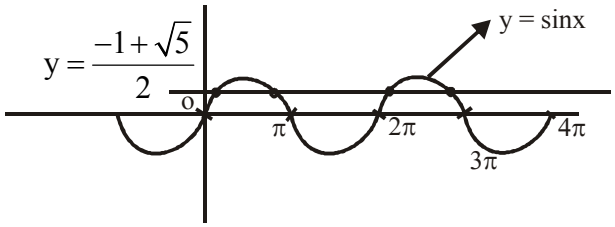
$\Rightarrow \sin x = 1 - \sin^2 x$

$\Rightarrow \sin^2 x + \sin x - 1 = 0$

$$\Rightarrow \sin x = \frac{-1 \pm \sqrt{1 - 4(-1)(1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

$\sin x \neq \frac{-1-\sqrt{5}}{2}$ $\quad \therefore \sin x \in [-1,1]$

So, $\sin x = \frac{\sqrt{5}-1}{2} > 0$



So total number of solution =4

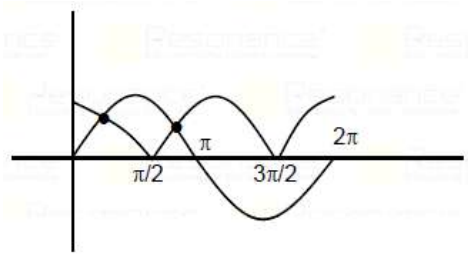
13. The number of solutions of $|\cos x| = \sin x$, $-4\pi \le x \le 4\pi$ is : **[JEE(Main)July2022]**

समीकरण $|\cos x| = \sin x$, $-4\pi \le x \le 4\pi$ के हलों की संख्या है : **[JEE(Main)July2022]**

(1) 4 (2) 6 (3) 8 (4) 12

Ans. Official Answer NTA (3)

Sol.



$\because$ |cos x| is periodic will period π and sinx is periodic with 2π
and as shown in figure |cosx| = sinx is having 2 solutions
$\therefore$ Total number of solution of the equation |cosx| = sinx is $[-4\pi, 4\pi]$ is equal to 4 5 2 = 8

14. Let $S = \{\theta \in [0, 2\pi] : 8^{2\sin^2\theta} + 8^{2\cos^2\theta} = 16\}$. Then $n(S) + \sum_{\theta \in S}\left(\sec\left(\frac{\pi}{4}+2\theta\right)\operatorname{cosec}\left(\frac{\pi}{4}+2\theta\right)\right)$ is equal to :

[JEE(Main)July2022]

माना $S = \{\theta \in [0, 2\pi] : 8^{2\sin^2\theta} + 8^{2\cos^2\theta} = 16\}$ है। तो $n(S) + \sum_{\theta \in S}\left(\sec\left(\frac{\pi}{4}+2\theta\right)\operatorname{cosec}\left(\frac{\pi}{4}+2\theta\right)\right)$ बराबर है :

[JEE(Main)July2022]

(1) 0 (2) – 2 (3) – 4 (4) 12

Ans. Official Answer NTA (3)

Sol. $8^{2\sin^2\theta} + 8^{2-2\sin^2\theta} = 16$

$$y+\frac{64}{y}=16$$

$$\Rightarrow y=8$$

$$\Rightarrow \sin^2\theta=1/2$$

$$n(S)+\sum_{\theta\in S}\frac{1}{\cos(\pi/4+2\theta)\sin(\pi/4+2\theta)}$$

$$=4+(-2)\times 4=-4$$

15. Let $S=\left\{\theta\in\left(0,\frac{\pi}{2}\right):\sum_{m=1}^{9}\sec\left(\theta+(m-1)\frac{\pi}{6}\right)\sec\left(\theta+\frac{m\pi}{6}\right)=-\frac{8}{\sqrt{3}}\right\}$. Then : **[JEE(Main)July2022]**

माना $S=\left\{\theta\in\left(0,\frac{\pi}{2}\right):\sum_{m=1}^{9}\sec\left(\theta+(m-1)\frac{\pi}{6}\right)\sec\left(\theta+\frac{m\pi}{6}\right)=-\frac{8}{\sqrt{3}}\right\}$ है। तो : **[JEE(Main)July2022]**

(1) $S=\left\{\frac{\pi}{12}\right\}$ (2) $S=\left\{\frac{2\pi}{3}\right\}$

(3) $\sum_{\theta\in S}\theta=\frac{\pi}{2}$ (4) $\sum_{\theta\in S}\theta=\frac{3\pi}{4}$

Ans. Official Answer NTA (3)

Sol. Let $\alpha=\theta+(m-1)\frac{\pi}{6}$

& $\beta=\theta+m\frac{\pi}{6}$

So, $\beta-\alpha=\frac{\pi}{6}$

Here, $\sum_{m=1}^{9}\sec\alpha.\sec\beta=\sum_{m=1}^{9}\frac{1}{\cos\alpha.\cos\beta}$

$$=2\sum_{m=1}^{9}\frac{\sin(\beta-\alpha)}{\cos\alpha.\cos\beta}=2\sum_{m=1}^{9}(\tan\beta-\tan\alpha)$$

$$=2\sum_{m=1}^{9}\left(\tan\left(\theta+m\frac{\pi}{6}\right)-\tan\left(\theta+(m-1)\frac{\pi}{6}\right)\right)$$

$$=2\left(\tan\left(\theta+\frac{9\pi}{6}\right)-\tan\theta\right)=2(-\cot\theta-\tan\theta)=-\frac{8}{\sqrt{3}}$$

(Given)

$$\therefore \tan\theta+\cot\theta=\frac{4}{\sqrt{3}}$$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{3}} \text{ or } \sqrt{3}$

So, $S = \left\{\frac{\pi}{6}, \frac{\pi}{3}\right\}$

$\sum_{\theta \in S} \theta = \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$

16. The number of elements in the set $S\left\{x \in R : 2\cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}\right\}$ is : **[JEE(Main)July2022]**

समुच्चय $S\left\{x \in R : 2\cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}\right\}$ में अवयवों की संख्या है : **[JEE(Main)July2022]**

(1) 1 (2) 3 (3) 0 (4) infinite (अनंत)

Ans. Official Answer NTA (1)

Sol. $2\cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}$

L.H.S ≤ 2. & R.H.S. ≥ 2

Hence L.H.S = 2 & R.H.S = 2

$2\cos\left(\frac{x^2+x}{6}\right) = 2 4^x + 4^{-x} = 2$

Check x = 0 Possible hence only one solution.

17. If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____. **[JEE(Main)July2022]**

यदि समीकरण निकाय $2\sin^2\theta - \cos 2\theta = 0$ तथा $2\cos^2\theta + 3\sin\theta = 0$ के अंतराल $[0, 2\pi]$ में हलों का योगफल $k\pi$ है, तो k बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (3)

Sol. $2\sin^2\theta - \cos 2\theta = 0$

$2\sin^2\theta - (1 - 2\sin^2\theta) = 0$

$\Rightarrow \sin^2\theta = \left(\frac{1}{2}\right)^2$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

$2\cos^2\theta + 3\sin\theta = 0$

$\Rightarrow 2\sin^2\theta - 3\sin\theta - 2 = 0$

$\therefore \sin\theta = -\frac{1}{2}$

$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$

So, the common solution is

$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$

K = 3

18. Let $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$. Then the number of elements in the set $A = \left\{\theta \in S : \tan\theta\left(1+\sqrt{5}\tan(2\theta)\right) = \sqrt{5} - \tan(2\theta)\right\}$ is _____. **[JEE(Main)July2022]**

माना $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$ है। तो समुच्चय $A = \left\{\theta \in S : \tan\theta\left(1+\sqrt{5}\tan(2\theta)\right) = \sqrt{5} - \tan(2\theta)\right\}$ में अवयवों की संख्या है_____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (5)

Sol. $\tan\theta + \sqrt{5}\tan 2\theta \tan\theta = \sqrt{5} - \tan 2\theta$

$\tan 3\theta = \sqrt{5}$

$\theta = \frac{n\pi}{3} + \frac{\alpha}{3}; \tan\alpha = \sqrt{5}$

Five solution

19. Let $S = \{\theta \in (0, 2\pi) : 7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2\}$. Then, the sum of roots of all the equations $x^2 - 2(\tan^2\theta + \cot^2\theta)x + 6\sin^2\theta = 0, \theta \in S$, is _____. **[JEE(Main)July2022]**

माना $S = \{\theta \in (0, 2\pi) : 7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2\}$ है। सभी समीकरणों $x^2 - 2(\tan^2\theta + \cot^2\theta)x + 6\sin^2\theta = 0, \theta \in S$ के मूलों का योग है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (16)

Sol. $7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2$

$4\cos^2\theta + 3\cos^2\theta \cdot 2\cos^2 2\theta = 2$

$2(1 + \cos 2\theta) + 3\cos 2\theta - 2\cos^2 2\theta = 2$

$2\cos^2 2\theta - 5\cos^2\theta = 0$

$\cos^2\theta(2\cos^2\theta - 5) = 0$

$\cos^2\theta = 0$

$2\theta = (2n+1)\frac{\pi}{2}$

$\theta = (2n+1)\frac{\pi}{4}$

$S = \left\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right\}$

For all four values of θ

$x^2 - 2(\tan^2\theta + \cot^2\theta)x + 6\sin^2\theta = 0$

$\Rightarrow x^2 - 4x + 3 = 0$

Sum of roots of all four equations $= 4 \times 4 = 16$.

Exercise J - A

1. For $x \in (0, \pi)$ the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has : **[JEE(Adv.)2014]**

(A) infinitely many solutions (B) three solutions

(C) one solution (D) no solution

$x \in (0, \pi)$ के लिए समीकरण $\sin x + 2\sin 2x - \sin 3x = 3$ के : **[JEE(Adv.)2014]**

(A) अनन्त हल होगें (B) तीन हल होगें

(C) एक हल होगा (D) कोई हल नहीं होगा

Ans. D

Sol. $\sin x + 2\sin 2x - \sin 3x = 3.$

$\Rightarrow \sin x (1 + 2\cos x - 3 + 4\sin^2 x) = 3.$

$\Rightarrow (4\sin^2 x + 2\cos x - 2) = \dfrac{3}{\sin x}$

$\Rightarrow 2 - 4\cos^2 x + 2\cos x = \dfrac{3}{\sin x}$

$\Rightarrow \dfrac{9}{4} - \left(2\cos x - \dfrac{1}{2}\right)^2 = \dfrac{3}{\sin x}.$

L.H.S. $\le \dfrac{9}{4}$ R.H.S. $\ge 3.$

No solution

2. The numer of distinct solutions of the equation $\dfrac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is. **[JEE(Adv.)2015]**

अंतराल $[0, 2\pi]$ में समीकरण $\dfrac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ के विभिन्न हलों की संख्या है। **[JEE(Adv.)2015]**

Ans. 8

Sol. $\dfrac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$

$\Rightarrow \quad \dfrac{5}{4}\cos^2 2x + 1 - \dfrac{1}{2}\sin^2 2x + 1 - \dfrac{3}{4}\sin^2 2x = 2$

$\Rightarrow \quad \cos^2 2x = \sin^2 2x$

$\Rightarrow \quad \tan^2 2x = 1$

$\because \quad 2x \in [0, 4\pi] \quad \Rightarrow \quad x = \dfrac{\pi}{8}, \dfrac{3\pi}{8}, \dfrac{5\pi}{8}, \dfrac{7\pi}{8}, \dfrac{9\pi}{8}, \dfrac{11\pi}{8}, \dfrac{13\pi}{8}, \dfrac{15\pi}{8}$

so number of solution is 8

3. Let $S=\left\{x\in(-\pi,\pi):x\neq 0,\pm\frac{\pi}{2}\right\}$. The sum of all distinct solutions of the equation $\sqrt{3}\sec x+\operatorname{cosec}x+2(\tan x-\cot x)=0$ in the set S is equal to : **[JEE(Adv.)2016]**

माना कि $S=\left\{x\in(-\pi,\pi):x\neq 0,\pm\frac{\pi}{2}\right\}$ है। समुच्चय S में समीकरण $\sqrt{3}\sec x+\operatorname{cosec}x+2(\tan x-\cot x)=0$ के सभी भिन्न हलों का योग है : **[JEE(Adv.)2016]**

(A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

Ans. C

Sol. $2\left(\frac{\sqrt{3}/2}{\cos x}+\frac{1/2}{\sin x}\right)+2\left(\tan x-\cot x\right)=0$

$\Rightarrow 2\frac{\cos(x-60^\circ)}{\sin x\cos x}+\frac{2(-\cos 2x)}{\sin x\cos x}=0$

$\Rightarrow \cos(x-60^\circ)=\cos 2x$

$\Rightarrow x=2n\pi-\frac{\pi}{3},\ (6m+1)\frac{\pi}{9}\ (m,n\in I)$

$\Rightarrow$ Solutions belonging to set $S=-\frac{\pi}{3},-\frac{5\pi}{9},\frac{\pi}{9},\frac{7\pi}{9}$

sum = 0

4. Let a, b, c be three non-zero real numbers such that the equation $\sqrt{3}a\cos x+2b\sin x=c, x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, has two distinct real roots α and β with $\alpha+\beta=\pi/3$. Then, the value of b/a is _____. **[JEE(Adv.)2018]**

माना कि a, b, c ऐसी तीन शून्येत्तर वास्तविक संख्याऐं हैं जिनके लिये समीकरण $\sqrt{3}a\cos x+2b\sin x=c, x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, के दो भिन्न वास्तविक मूल α और β हैं, जहाँ $\alpha+\beta=\pi/3$. तब b/a का मान है। **[JEE(Adv.)2018]**

Ans. 0.5

Sol. $\sqrt{3}\ a\cos x+2b\sin x=c \quad x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$\Rightarrow \sqrt{3}\ a\left(\frac{1-t^2}{1+t^2}\right)+2b\left(\frac{2t}{1+t^2}\right)=c,$ where $t=\tan\frac{x}{2}$

$\Rightarrow \sqrt{3}\,a(1-t^2)+4bt=c(1+t^2)$

$\Rightarrow t^2(c+\sqrt{3}a)-4bt+c-\sqrt{3}a=0$

$\because \quad \frac{\alpha+\beta}{2}=\frac{\pi}{6}$

$\Rightarrow \tan\left(\frac{\alpha+\beta}{2}\right)=\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{t_1+t_2}{1-t_1t_2}=\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{4b}{c+\sqrt{3}a-c+\sqrt{3}a}=\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{b}{a}=\frac{1}{2}$

5. Consider the following column : **[JEE(Adv.)2022]**

Column - I	**Column - II**
(I) $\left\{x\in\left[-\frac{2\pi}{3},\frac{2\pi}{3}\right]:\cos x+\sin x=1\right\}$	(P) has two elements
(II) $\left\{x\in\left[-\frac{5\pi}{18},\frac{5\pi}{18}\right]:\sqrt{3}\tan 3x=1\right\}$	(Q) has three elements
(III) $\left\{x\in\left[-\frac{6\pi}{5},\frac{6\pi}{5}\right]:2\cos(2x)=\sqrt{3}\right\}$	(R) has four elements
(IV) $\left\{x\in\left[-\frac{7\pi}{4},\frac{7\pi}{4}\right]:\sin x-\cos x=1\right\}$	(S) has five elements
	(T) has six elements

The correct option is:

(A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)

(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

निम्न कॉलम पर विचार कीजिए : **[JEE(Adv.)2022]**

Column - I	**Column - II**
(I) $\left\{x\in\left[-\frac{2\pi}{3},\frac{2\pi}{3}\right]:\cos x+\sin x=1\right\}$	(P) में दो अवयव है
(II) $\left\{x\in\left[-\frac{5\pi}{18},\frac{5\pi}{18}\right]:\sqrt{3}\tan 3x=1\right\}$	(Q) में तीन अवयव है

(III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$ (R) में चार अवयव है

(IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$ (S) में पांच अवयव है

(T) में छः अवयव है

सही विकल्प है :

(A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)

(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

Ans. (B)

Sol. $\left\{x \in \left[\frac{-2\pi}{3}, \frac{2\pi}{3}\right], \cos x + \sin x = 1\right\}$

$\cos x + \sin x = 1$

$\sin\left(\frac{\pi}{4} + x\right) = \frac{1}{\sqrt{2}}$

$\frac{\pi}{4} + x = n\pi + (-1)^n \frac{\pi}{4}$

$x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$

$\therefore$ x has 2 elements. $\rightarrow$ P

(ii) $\left\{x \in \left[\frac{-5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$

$\sqrt{3}\tan 3x = 1$

$\tan 3x = \frac{1}{\sqrt{3}}$

$3x = n\pi + \frac{\pi}{6}$

$x = \frac{n\pi}{3} + \frac{\pi}{18}$

$\therefore$ x has 2 elements. $\rightarrow$ P

(iii) $\left\{x \in \left[\frac{-6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos 2x = \sqrt{3}\right\}$

$2\cos 2x = \sqrt{3}$

$$\cos 2x = \frac{\sqrt{3}}{2}$$

$$2x = 2n\pi \pm \frac{\pi}{6}$$

$$x = n\pi \pm \frac{\pi}{12}$$

$\therefore$ x has 6 elements. $\rightarrow$ T

(iv) $\left\{ x \in \left[\frac{-7\pi}{4}, \frac{7\pi}{4} \right] : \sin x - \cos x = 1 \right\}$

$\sin x - \cos x = 1$

$$\sin\left(x - \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$x - \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$$

$$x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}$$

$\therefore$ x has 4 elements. $\rightarrow$ R

$\therefore$ option B is correct.

CHALLENGER EXERCISES

Exercise C - I

1. If $\tan(\pi\cos\theta) = \cot(\pi\sin\theta)$, then the value(s) of $\cos\left(\theta - \frac{\pi}{4}\right)$ is (are) :

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\pm\frac{1}{2\sqrt{2}}$ (D) None of these

यदि $\tan(\pi\cos\theta) = \cot(\pi\sin\theta)$ तब $\cos\left(\theta - \frac{\pi}{4}\right)$ का मान ज्ञात होगा :

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\pm\frac{1}{2\sqrt{2}}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\tan(\pi\cos\theta) = \cot(\pi\sin\theta)$

$$\Rightarrow \tan(\pi\cos\theta) = \tan\left(\frac{\pi}{2} - \pi\sin\theta\right)$$

$$\Rightarrow \pi\cos\theta = n\pi + \frac{\pi}{2} - \pi\sin\theta$$

$$\Rightarrow \cos\theta + \sin\theta = \frac{(2n+1)}{2} \Rightarrow \cos\left(\theta - \frac{\pi}{4}\right) = \left(\frac{2n+1}{2\sqrt{2}}\right)$$

For n = 0 $\cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$

For n = 1 $\cos\left(\theta - \frac{\pi}{4}\right) = \frac{3}{2\sqrt{2}}$

For n = –1 $\cos\left(\theta - \frac{\pi}{4}\right) = \frac{-1}{2\sqrt{2}}$

2. The equation , $\sin^4 x - 2\cos^2 x + a^2 = 0$ is solvable if :

(A) $-\sqrt{3} \le a \le \sqrt{3}$ (B) $-\sqrt{2} \le a \le \sqrt{2}$ (C) $-1 \le a \le 1$ (D) None of these

समीकरण $\sin^4 x - 2\cos^2 x + a^2 = 0$ का हल संभव है यदि :

(A) $-\sqrt{3} \le a \le \sqrt{3}$ (B) $-\sqrt{2} \le a \le \sqrt{2}$ (C) $-1 \le a \le 1$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\sin^4x - 2\cos^2x + a^2 = 0$

$\Rightarrow \sin^4x - 2(1 - \sin^2x) + a^2 = 0$

$\Rightarrow \sin^4x + 2\sin^2x + (a^2 - 2) = 0$

$\Rightarrow (\sin^2x + 1)^2 + a^2 - 3 = 0$

$\Rightarrow a^2 = 3 - (\sin^2 x + 1)^2$

$\because \sin^2 x \in [0,1]$

$\Rightarrow (\sin^2 x + 1) \in [1, 2]$

$\Rightarrow (\sin^2 x + 1)^2 \in [1, 4]$

$\Rightarrow 3 - (\sin^2 x + 1)^2 \in [-1, 2]$

$\Rightarrow 0 \le a^2 \le 2 \qquad \Rightarrow -\sqrt{2} \le a \le \sqrt{2}$

3. The general solution of the equation , $\dfrac{1 - \sin x + ... + (-1)^n \sin^n x + ...}{1 + \sin x + ... + \sin^n x + ...} = \dfrac{1 - \cos 2x}{1 + \cos 2x}$ is :

निम्न समीकरण $\dfrac{1 - \sin x + ... + (-1)^n \sin^n x + ...}{1 + \sin x + ... + \sin^n x + ...} = \dfrac{1 - \cos 2x}{1 + \cos 2x}$ का व्यापक हल है :

(A) $(-1)^n \left(\dfrac{\pi}{3}\right) + n\pi$ (B) $(-1)^n \left(\dfrac{\pi}{6}\right) + n\pi$ (C) $(-1)^{n+1} \left(\dfrac{\pi}{6}\right) + n\pi$ (D) $(-1)^{n-1} \left(\dfrac{\pi}{3}\right) + n\pi$, $(n \in I)$

Ans. B

Sol. $\dfrac{1 - \sin x + ... + (-1)^n \sin^n x + ...}{1 + \sin x + ... + \sin^n x + ...} = \dfrac{1 - \cos 2x}{1 + \cos 2x}$

$\Rightarrow \dfrac{\dfrac{1}{1 + \sin x}}{\dfrac{1}{1 - \sin x}} = \tan^2 x$

$\Rightarrow \dfrac{1 - \sin x}{1 + \sin x} = \tan^2 x$

$\Rightarrow (1 - \sin^2 x)(1 - \sin x) = (1 + \sin x)\sin^2 x$

$\Rightarrow 2\sin^2 x + \sin x - 1 = 0$

$\Rightarrow (2\sin x - 1)(\sin x + 1) = 0$

$\Rightarrow \underbrace{\sin x = -1}_{\text{Not possible as } \cos x \neq 0}$ or $\sin x = \dfrac{1}{2}$

$\Rightarrow \sin x = \dfrac{1}{2}$

$\Rightarrow x = n\pi + (-1)^n \dfrac{\pi}{6}; n \in I$

4. If the equation , $x^2 + 4 + 3\sin(ax + b) - 2x = 0$ has atleast one real solution, where $a, b \in [0, 2\pi]$, then one possible value of $(a + b)$ can be equal to :

(A) $\dfrac{7\pi}{2}$ (B) $\dfrac{5\pi}{2}$ (C) $\dfrac{9\pi}{2}$ (D) None of these

यदि समीकरण $x^2 + 4 + 3\sin(ax + b) - 2x = 0$ के कम से कम एक वास्तविक हल है जहां $a, b \in [0, 2\pi]$ तब $(a + b)$ का एक सम्भव हल होगा :

(A) $\frac{7\pi}{2}$ (B) $\frac{5\pi}{2}$ (C) $\frac{9\pi}{2}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $x^2 + 4 + 3\sin(ax + b) - 2x = 0$

$$\Rightarrow \frac{1}{3}\left[-x^2 + 2x - 4\right] = \sin(ax + b)$$

$\because \sin(ax + b) \in [-1, 1]$

$$\& \quad -\frac{x^2}{3} + \frac{2x}{3} - \frac{4}{3} = 0 \in \left(-\infty, -\frac{D}{4a}\right] \equiv (-\infty, -1]$$

Only possibility are $\underbrace{\sin(ax + b) = -1}_{\text{only for } x = 1}$

$$\Rightarrow a + b = (4n - 1)\frac{\pi}{2}$$

$$\Rightarrow a + b = \frac{7\pi}{2} \qquad \{\because a, b \in [0, 2\pi]\}$$

5. The number of points of intersection of the two curves, $y = 2\sin x$ and $y = 5x^2 + 2x + 3$, is :

(A) Infinite (B) 0 (C) 1 (D) 2

दो वक्रो $y = 2\sin x$ एवं $y = 5x^2 + 2x + 3$ के प्रतिच्छेदन से बने बिन्दुओं की संख्या है :

(A) अन्नत (B) 0 (C) 1 (D) 2

Ans. B

Sol. $\underbrace{y = 2\sin x}_{\because 2\sin x \in [-2,2]}$ & $y = 5x^2 + 2x + 3$

$D = 4 - 60 = -56 < 0$

$a = 5 > 0$

$$-\frac{b}{2a} = \frac{-2}{10} = \frac{-1}{5}$$

$$-\frac{D}{4a} = \frac{56}{4 \times 5} = \frac{14}{5}$$

$$5x^2 + 2x + 3 \in \left[\frac{14}{5}, \infty\right]$$

$\because$ No. of points of intersection of the two curve is 0.

6. The number of all possible triplets (a_1, a_2, a_3) such that $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0$ for all x is :

(A) zero (B) one (C) three (D) inifnite

सभी x के लिए समीकरण $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0$ त्रिक (a_1, a_2, a_3) की संख्या होगी :

(A) शून्य (B) एक (C) तीन (D) अन्नत

Ans. D

Sol. $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0 \quad \forall x \in R$

$\Rightarrow a_1 + a_2 + \sin^2 x\,(a_3 - 2a_2) = 0 \quad \forall x \in R$

$\Rightarrow a_1 + a_2 = 0 \quad \& \quad a_3 - 2a_2 = 0$

$\Rightarrow (a_1, a_2, a_3) = (-a_2, a_2, 2a_2) = a_2(-1, 1, 2)$

$\therefore$ There are infinite triplets.

7. The number of values of x in the interval $(0, 5\pi)$, satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is :

अन्तराल $(0, 5\pi)$ में समीकरण $3\sin^2 x - 7\sin x + 2 = 0$ के हलों की संख्या है :

(A) 0 (B) 5 (C) 6 (D) 10

Ans. C

Sol. Given $x \in (0, 5\pi)$

$3\sin^2 x - 7\sin x + 2 = 0$

$\Rightarrow (3\sin x - 1)(\sin x - 2) = 0$

$\Rightarrow \underbrace{\sin x = 2}_{\text{Not possible}}$ or $\sin x = \frac{1}{3}$

$\Rightarrow \sin x = \frac{1}{3}$; In $(0, 5\pi)$ No. of values of x is 6.

8. If $\sin^2\theta - \sin\theta - 1 = 0$ is to be satisficed for exactly 4 distinct value of $\theta \in [0, n\pi]$, $n \in N$, then least value of n is:

यदि समीकरण $\sin^2\theta - \sin\theta - 1 = 0$ चार अलग–अलग मानों के लिए सन्तुष्ट होती है तथा $\theta \in [0, n\pi]$, $n \in N$, तब n का न्यूनतम मान होगा :

(A) 2 (B) 6 (C) 4 (D) 1

Ans. C

Sol. $\sin^2\theta - \sin\theta - 1 = 0$

$\Rightarrow \sin\theta = \frac{1 \pm \sqrt{5}}{2}$

$\Rightarrow \underbrace{\sin\theta = \frac{1+\sqrt{5}}{2}}_{\text{Not possible}}$ or $\sin\theta = \frac{1-\sqrt{5}}{2}$

In $[0, 4\pi]$ No. of distinct values of θ is 4.

$\therefore n = 4$

9. If $\tan(\cot x) = \cot(\tan x)$, then $\sin 2x =$

यदि $\tan(\cot x) = \cot(\tan x)$ तब $\sin 2x =$

(A) $\frac{4}{(2n+1)\pi}; n \in I$ (B) $\frac{4}{(2n+1)\pi}; n \in I - \{-1, 0\}$

(C) $\frac{4}{(2n+1)\pi}; n \in I - \{-1\}$ (D) $\frac{4}{(2n+1)\pi}; n \in I - \{0\}$

Ans. B

Sol. $\tan(\cot x) = \cot(\tan x)$

$\Rightarrow \tan(\cot x) = \tan\left(\frac{\pi}{2} - \tan x\right)$

$\Rightarrow \cot x = n\pi + \frac{\pi}{2} - \tan x$

$\Rightarrow \tan x + \cot x = n\pi + \frac{\pi}{2}$

$\Rightarrow \frac{2}{\sin 2x} = (2n+1)\frac{\pi}{2}$

$\Rightarrow \sin 2x = \frac{4}{(2n+1)\pi}; n \in Z - \{-1, 0\}$

10. The general solution of the equation $\tan^2\alpha + 2\sqrt{3}\tan\alpha = 1$ is given by :

समीकरण $\tan^2\alpha + 2\sqrt{3}\tan\alpha = 1$ का व्यापक हल है :

(A) $\alpha = \frac{n\pi}{2}$ (B) $\alpha = (2n+1)\frac{\pi}{2}$ (C) $\alpha = (6n+1)\frac{\pi}{12}$ (D) $\alpha = n\frac{\pi}{12}$

Ans. C

Sol. $\tan^2\alpha + 2\sqrt{3}\tan\alpha - 1 = 0$

$\tan\alpha = \frac{-2\sqrt{3} \pm 4}{2}$

$\Rightarrow \tan\alpha = -(2+\sqrt{3})$ or $\tan\alpha = 2 - \sqrt{3}$

$\Rightarrow \tan\alpha = \tan\left(\frac{\pi}{2} + \frac{\pi}{12}\right)$ or $\tan\alpha = \tan\left(\frac{\pi}{12}\right)$

$\Rightarrow \alpha = n\pi + \frac{7\pi}{12}$ or $\alpha = n\pi + \frac{\pi}{12}$

$\Rightarrow \alpha = (6n+1)\frac{\pi}{12}$

11. Number of real values of x in $[-2\pi, 2\pi]$ for which value of the expression $y = \sqrt{2\sin x} - \sin x$ is maximum, is :

(A) 0 (B) 2 (C) 4 (D) more than 4

अन्तराल $[-2\pi, 2\pi]$ में x के उन वास्तविक मानों की संख्या, जिनके लिए व्यंजक $y = \sqrt{2\sin x} - \sin x$ का मान अधिकतम है, होगी :

(A) 0 (B) 2 (C) 4 (D) 4 से अधिक

Ans. C

Sol. $x \in [-2\pi, 2\pi]$

$y = \sqrt{2\sin x} - \sin x$

Let $\sqrt{\sin x} = t$

$\Rightarrow \sin x = t^2$

$\Rightarrow y = \sqrt{2}t - t^2$

y is maximum at Abscissa value of vertex.

$\because$ Abscissa value of vertex $= \frac{-b}{2a} = \frac{-\sqrt{2}}{-2\times 1} = \frac{1}{\sqrt{2}}$

$\Rightarrow t = \frac{1}{\sqrt{2}}$

$\Rightarrow \sqrt{\sin x} = \frac{1}{\sqrt{2}} \Rightarrow \sin x = \frac{1}{2}$

In $[-2\pi, 2\pi]$ no. of real values of x is 4.

$\therefore n = 4$

12. The set of all x in $(-\pi, \pi)$ satisfying $|4\sin x - 1| < \sqrt{5}$ is given by :

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (B) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (C) $\left(\frac{\pi}{10}, -\frac{3\pi}{10}\right)$ (D) none of these

अन्तराल $(-\pi, \pi)$ में असमिका $|4\sin x - 1| < \sqrt{5}$ को संतुष्ट करने वाले x के सभी मानों का समुच्चय होगा :

(A) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (B) $\left(\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (C) $\left(\frac{\pi}{10}, -\frac{3\pi}{10}\right)$ (D) इनमें से कोई नहीं

Ans. A

Sol. $|4\sin x - 1| < \sqrt{5} \quad \Rightarrow \quad -\sqrt{5} < 4\sin x - 1 < \sqrt{5}$

$$\Rightarrow \quad -\left(\frac{\sqrt{5}-1}{4}\right) < \sin x < \left(\frac{\sqrt{5}+1}{4}\right)$$

$$\Rightarrow \quad -\sin\frac{\pi}{10} < \sin x < \sin\frac{3\pi}{10}$$

$$\Rightarrow \quad \sin\left(-\frac{\pi}{10}\right) < \sin x < \sin\frac{3\pi}{10}$$

$$\Rightarrow x \in \left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$$

13. The values of x between 0 and 2π, which satisfy the equation $\sin x\sqrt{8\cos^2 x} = 1$ are in A.P. with common difference:

0 से 2π के मध्य x के मान, जो समीकरण $\sin x\sqrt{8\cos^2 x} = 1$ को संतुष्ट करते है, समान्तर श्रेणी में है तो सार्वअन्तर होगा :

(A) $\pi/4$ (B) $\pi/8$ (C) $3\pi/8$ (D) $5\pi/8$

Ans. A

Sol. $x \in (0, 2\pi)$

$$\sin x\sqrt{8\cos^2 x} = 1$$

$$2\sin x|\cos x| = \frac{1}{\sqrt{2}}$$

Case I :- If $\cos x \geq 0$

then $\sin 2x = \frac{1}{\sqrt{2}}$

$$\Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{8}$$

$\because \cos x > 0$

$$\Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}$$

Case II :- If $\cos x < 0$ then $\sin 2x = -\frac{1}{\sqrt{2}}$

$$\Rightarrow 2x = \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{13\pi}{4}, \frac{15\pi}{4}$$

$\because \cos x < 0$

$$\Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8}$$

$$\therefore x = \underbrace{\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}}_{\text{A.P. with common difference} = \frac{\pi}{4}}$$

14. α, β, γ and δ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k . The value of $4 \sin \frac{\alpha}{2} + 3 \sin \frac{\beta}{2} + 2 \sin \frac{\gamma}{2} + \sin \frac{\delta}{2}$ is equal to :

यदि α, β, γ तथा δ बढ़ते हुये क्रम में न्यूनतम धनात्मक कोण हो जहाँ सभी के sines का मान समान है और k के बराबर है तो $4 \sin \frac{\alpha}{2} + 3 \sin \frac{\beta}{2} + 2 \sin \frac{\gamma}{2} + \sin \frac{\delta}{2}$ का मान होगा :

(A) $2\sqrt{1-k}$ (B) $2\sqrt{1+k}$ (C) $2\sqrt{k}$ (D) $2k$

Ans. B

Sol. $\because \alpha < \beta < \gamma < \delta$ and $\sin\alpha = \sin\beta = \sin\gamma = \sin\delta = k$

$\Rightarrow \alpha, \beta = \pi - \alpha$, $\gamma = 2\pi + \alpha$, $\delta = 3\pi - \alpha$

$$\Rightarrow y = 4\sin\frac{\alpha}{2} + 3\sin\frac{\beta}{2} + 2\sin\frac{\gamma}{2} + \sin\frac{\delta}{2}$$

$$\Rightarrow y = 4\sin\frac{\alpha}{2} + 3\sin\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) + 2\sin\left(\pi + \frac{\alpha}{2}\right) + \sin\left(\frac{3\pi}{2} - \frac{\alpha}{2}\right)$$

$$\Rightarrow y = 4\sin\frac{\alpha}{2} + 3\cos\frac{\alpha}{2} - 2\sin\frac{\alpha}{2} - \cos\frac{\alpha}{2}$$

$$\Rightarrow y = 2\left(\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right)$$

$$\Rightarrow y^2 = 4(1 + \sin\alpha)$$

$\Rightarrow y^2 = 4(1 + k)$

$\Rightarrow y = 2\sqrt{1+k}$

15. The number of points inside or on the circle $x^2 + y^2 = 4$ satisfying $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ is :

(A) one (B) two (C) four (D) infinite

वृत $x^2 + y^2 = 4$ पर या अन्दर स्थित बिन्दुओं की संख्या, जो $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$ को संतुष्ट करते है :

(A) एक (B) दो (C) चार (D) अनन्त

Ans. C

Sol. $x^2 + y^2 = 4$ & $\tan^4 x + \cot^4 x + 1 = 3 \sin^2 y$

$\because \tan^4 x + \cot^4 x + 1 \in [3, \infty)$

& $3 \sin^2 y \in [0, 3]$

$\Rightarrow 3\sin^2 y = 3$ & $\tan^4 x + \cot^4 x + 1 = 3$

$\Rightarrow \sin^2 y = 1$ & $\tan^4 x + \cot^4 x = 2$

$\Rightarrow y = (2n+1)\frac{\pi}{2}$ & $\tan^4 x = 1$

$\Rightarrow y = (2n+1)\frac{\pi}{2}$ & $x = n\pi \pm \frac{\pi}{4}$

$\Rightarrow y = -\frac{\pi}{2}, \frac{\pi}{2}$ & $x = \frac{\pi}{4}, -\frac{\pi}{4}$ $\left\{\because x^2 + y^2 = 4\right\}$

$\Rightarrow$ No. of pairs are 4.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

16. $|\cos x| = \cos x - 2\sin x$, if :

$|\cos x| = \cos x - 2\sin x$, यदि :

(A) $x = n\pi$ (B) $x = 2n\pi$ (C) $x = n\pi + \frac{\pi}{4}$ (D) $x = (2n+1)\pi + \frac{\pi}{4}$

Ans. BD

Sol. $|\cos x| = \cos x - 2\sin x$

Case I :- If $\cos x \geq 0$ then

$\sin x = 0$

$\Rightarrow x = n\pi;\ n \in I$

$\Rightarrow x = 0, \pi, 2\pi, 3\pi, ...$

$\because \cos x \geq 0$

$\Rightarrow x = 0, 2\pi, 4\pi, ...$

$\Rightarrow x = 2n\pi;\ n \in I$

Case II :- If $\cos x < 0$

then $\tan x = 1$

$\Rightarrow x = n\pi + \frac{\pi}{4}$

$\because \cos x < 0$

$\Rightarrow x = (2n+1)\pi + \frac{\pi}{4};\ n \in I$

17. Consider the system of equations $\sin x \cos 2y = (a^2-1)^2 + 1$ and $\cos x \sin 2y = a + 1$. Which of the following ordered pairs (x, y) of real numbers can satisfy the given system of equations for permissible real values of a :

समीकरणों $\sin x \cos 2y = (a^2-1)^2 + 1$ तथा $\cos x \sin 2y = a + 1$ के निकाय पर ध्यान दें, वास्तविक संख्याओं का क्रमित युग्म (x, y) दिये गये समीकरणों के निकाय को a के सभी संभव मानों के लिए संतुष्ट करता है, होगा :

(A) $\left(\frac{-\pi}{2},\frac{-\pi}{2}\right)$ (B) $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$ (C) $\left(\frac{3\pi}{2},\frac{-\pi}{2}\right)$ (D) $\left(\frac{-\pi}{2},\frac{3\pi}{2}\right)$

Ans. ACD

Sol. $\because \sin x \cos 2y = (a^2-1)^2 + 1$

$\because \sin x \cos 2y \in [-1, 1]$

& $(a^2-1)^2 + 1 \in [1, \infty)$

$\Rightarrow (a^2-1)^2 + 1 = 1$

$\Rightarrow a^2 - 1 = 0$

$\Rightarrow a = \pm 1$

$\Rightarrow \sin x \cos 2y = 1$

$\Rightarrow \sin x = 1$ & $\cos 2y = 1$ or $\sin x = -1$ & $\cos 2y = -1$

$\Rightarrow x = (4n+1)\frac{\pi}{2}$ & $y = n\pi$ or $x = 2n\pi - \frac{\pi}{2}$ & $y = (2n+1)\frac{\pi}{2}$

$\Rightarrow (x, y) = \left(-\frac{\pi}{2},\frac{\pi}{2}\right),\left(-\frac{\pi}{2},-\frac{\pi}{2}\right),\left(\frac{3\pi}{2},-\frac{\pi}{2}\right),\left(-\frac{\pi}{2},\frac{3\pi}{2}\right)$

18. If $(\cos^2 x + \sec^2 x)(1 + \tan^2 2y)(3 + \sin 3z) = 4$, then :

(A) x may be a multiple of π (B) x cannot be an even multiple of π

(C) z can be a multiple of π (D) y can be a multiple of $\frac{\pi}{2}$

यदि $(\cos^2 x + \sec^2 x)(1 + \tan^2 2y)(3 + \sin 3z) = 4$, तो :

(A) x, π का गुणक हो सकता है (B) x, π का सम गुणक नहीं हो सकता

(C) z, π का गुणक हो सकता है (D) y, $\frac{\pi}{2}$ का गुणक हो सकता है

Ans. AD

Sol. $\underbrace{\left(\cos^2 x + \frac{1}{\cos^2 x}\right)}_{\in[2,\infty)}\underbrace{(1+\tan^2 2y)}_{[1,+\infty)}\underbrace{(3+\sin 3z)}_{\in[2,4]} = 4$

Only possibility are

$\cos^2 x + \frac{1}{\cos^2 x} = 2 \quad \Rightarrow x = n\pi;\ n \in I$

& $\sec^2 2y = 1 \quad \Rightarrow y = \frac{n\pi}{2};\ n \in I$

& $3 + \sin 3z = 2$

$\Rightarrow \sin 3z = -1$

$\Rightarrow z = \frac{n\pi}{3} + (-1)^n \frac{\pi}{2};\ n \in I$

19. A solution (x, y) of the system of equations $x - y = 1/3$ and $\cos^2(\pi x) - \sin^2(\pi y) = 1/2$ is given by :
समीकरण निकाय $x - y = 1/3$ तथा $\cos^2(\pi x) - \sin^2(\pi y) = 1/2$ का हल (x, y) दिया जा सकता है :
(A) (7/6, 5/6) (B) (2/3, 1/3) (C) (–5/6, –7/6) (D) (13/6, 11/6)

Ans. ACD

Sol. $x - y = \frac{1}{3}$ ...(i)

$\cos^2 \pi x - \sin^2(\pi y) = \frac{1}{2}$

$\Rightarrow \cos \pi(x + y) \cdot \sin \pi(x - y) = \frac{1}{2}$

$\Rightarrow \cos \pi(x + y) \sin \frac{\pi}{3} = \frac{1}{2}$ $\left\{\because x - y = \frac{1}{3}\right\}$

$\Rightarrow \cos \pi (x + y) = 1$

$\Rightarrow x + y = \frac{2n\pi}{\pi}; n \in I$

$\Rightarrow x + y = 2n$ ...(ii)

from (i) & (ii)

$x = n + \frac{1}{6}$ & $y = n - \frac{1}{6}$

For $n = -1$ $x = -\frac{5}{6}$ $y = -\frac{7}{6}$

$n = 0$ $x = \frac{1}{6}$ $y = -\frac{1}{6}$

$n = 1$ $x = \frac{7}{6}$ $y = \frac{5}{6}$

$n = 2$ $x = \frac{13}{6}$ $y = \frac{11}{6}$

20. $\tan|x| = |\tan x|$ if :
$\tan|x| = |\tan x|$ यदि :

(A) $x \in \left(-\frac{\pi(2k+1)}{2}, -\pi k\right]$ (B) $x \in \left[\pi k, \frac{\pi(2k+1)}{2}\right)$

(C) $x \in \left(-\pi k, -\frac{\pi(2k-1)}{2}\right)$ (D) $x \in \left(\frac{\pi(2k-1)}{2}, \pi k\right), k \in N$

Ans. AB

Sol. $\tan|x| = |\tan x|$

Case I :- If $x \in [0, \infty)$

$\Rightarrow \tan x = |\tan x|$

If $\tan x \geq 0$, then $\underbrace{\tan x = \tan x}_{\text{always true}}$

If $\tan x < 0$ then $\underbrace{\tan x = 0}_{\text{Not possible}}$

$\Rightarrow x \in \left[k\pi, (2k+1)\frac{\pi}{2}\right)$

Case II :- If $x \in (-\infty, 0)$

$-\tan x = |\tan x|$

If $\tan x > 0$ then $\underbrace{\tan x = 0}_{\text{Not possible}}$

If $\tan x \leq 0$ then $-\tan x = -\tan x$

$\Rightarrow x \in \left(-(2k+1)\frac{\pi}{2}, -\pi k\right)$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Find the coordinates of the points of intersection of the curves $y = \cos x$, $y = \sin 3x$ if $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.

वक्रों $y = \cos x$ व $y = \sin 3x$ के प्रतिच्छेदन से बने बिन्दुओं के निर्देशांक ज्ञात करे यदि $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.

Ans. $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right), \left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right), \left(-\frac{3\pi}{8}, \cos\frac{3\pi}{8}\right)$

Sol. $y = \cos x; y = \sin 3x$

$\Rightarrow$ P.O.I. are

$\cos x = \sin 3x$

$\Rightarrow \cos\left(\frac{\pi}{2} - 3x\right) = \cos x$

$\Rightarrow \frac{\pi}{2} - 3x = 2n\pi \pm x; n \in I$

$\Rightarrow \frac{\pi}{2} - 3x = 2n\pi + x; n \in I$ or $\frac{\pi}{2} - 3x = 2n\pi - x; n \in I$

$\Rightarrow 4x = \frac{\pi}{2} - 2n\pi; n \in I$ or $2x = \frac{\pi}{2} - 2n\pi; n \in I$

$\Rightarrow x = \frac{\pi}{8} - \frac{n\pi}{2}; n \in I$ or $x = \frac{\pi}{4} - n\pi; n \in I$

$\because x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\Rightarrow x = \frac{\pi}{8}, -\frac{3\pi}{8}, \frac{\pi}{4}$

$\Rightarrow y = \cos\frac{\pi}{8}, \cos\frac{3\pi}{8}, \frac{1}{\sqrt{2}}$

$\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right), \left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right), \left(-\frac{3\pi}{8}, \cos\frac{3\pi}{8}\right)$

2. Find the values of $x \in (-\pi, \pi)$ which satisfy the equation, $8^{\left(1 + |\cos x| + |\cos x|^2 + |\cos x|^3 + ...\right)} = 4^3$.

अन्तराल $x \in (-\pi, \pi)$ में x का मान ज्ञात करो जो निम्न समीकरण $8^{\left(1 + |\cos x| + |\cos x|^2 + |\cos x|^3 + ...\right)} = 4^3$ को सन्तुष्ट कर सके।

Ans. $\pm\frac{\pi}{3}$, $\pm\frac{2\pi}{3}$

Sol. Let $|\cos x| = t$

$\Rightarrow 8^{\left(1+|\cos x|+|\cos x|^2+.....\right)} = 4^3 = 2^6 \Rightarrow 8^{\left(1+t+t^2+.....\right)} = 2^6$

$\Rightarrow 8^{\frac{1}{1-t}} = 2^6$ $\qquad\qquad$ $\Rightarrow 2^{\frac{3}{1-t}} = 2^6$

$\Rightarrow \frac{3}{6} = 1-t$ $\qquad\qquad$ $\Rightarrow t = \frac{1}{2}$

$\Rightarrow |\cos x| = \frac{1}{2}$

$\because\ x \in (-\pi, \pi)$

$\Rightarrow x = \pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$

3. Solve the following system of equations for x and y

$5^{\left(\cos ec^2 x - 3\sec^2 y\right)} = 1$ and $2^{\left(2\cos ec\, x - \sqrt{3}\,|\sec y|\right)} = 64$

x व y के लिए निम्न समीकरण निकाय का हल ज्ञात करे

$5^{\left(\cos ec^2 x - 3\sec^2 y\right)} = 1$ and $2^{\left(2\cos ec\, x - \sqrt{3}\,|\sec y|\right)} = 64.$

Ans. $x = k\pi + (-1)^n\left(\sin^{-1}\frac{1}{6}\right)$, $y = t\,\pi \pm \left(\cos^{-1}\frac{1}{\sqrt{12}}\right)$

Sol. $5^{\cos ec^2 x - 3\sec^2 y} = 1$

$\Rightarrow \text{cosec}^2 x - 3\sec^2 y = 0$

$\Rightarrow \left(\cos ec x + \sqrt{3}\sec y\right)\left(\cos ec x - \sqrt{3}\sec y\right) = 0$

$\Rightarrow \cos ec x + \sqrt{3}\sec y = 0$ $\qquad$ or $\qquad$ $\cos ec x - \sqrt{3}\sec y = 0$

Now $2^{\left(2\cos ec x - \sqrt{3}|\sec y|\right)} = 64 = 2^6$

$\Rightarrow 2\cos ec x - \sqrt{3}\,|\sec y| = 6$

If $\sec y > 0$ then $2\cos ec x - \sqrt{3}\sec y = 6$

If $\sec y < 0$ then $2\cos ec x + \sqrt{3}\sec y = 6$

Case I :- If $\cos ec x - \sqrt{3}\sec y = 0$ then $\cos ec x = \sqrt{3}\sec y$

$\Rightarrow 2\sqrt{3}\sec y - \sqrt{3}\,|\sec y| = 6$

$\Rightarrow 2\sec y - |\sec y| = \frac{6}{\sqrt{3}} = 2\sqrt{3}$

$\Rightarrow \sec y = 2\sqrt{3}$ or $\sec y = \frac{2}{\sqrt{3}}$ (If $\sec y < 0$)

$\Rightarrow \sec y = \sqrt{12}$

$\Rightarrow \cos y = \frac{1}{\sqrt{12}}$

$\Rightarrow y = 2n\pi \pm \cos^{-1}\left(\frac{1}{\sqrt{12}}\right)$

& $x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{6}\right)$

Case II :- If $\text{cosec}\, x + \sqrt{3}\sec y = 0$ & $2\sqrt{3}\sec y - \sqrt{3}|\sec y| = 6$

If $\sec y > 0$ then $\sec y = -\frac{6}{3\sqrt{3}} < 0$ (Not possible)

If $\sec y < 0$ then $\sec y = -\sqrt{12}$

$\Rightarrow \cos y = -\frac{1}{\sqrt{12}}$

$\Rightarrow y = 2n\pi \pm \cos^{-1}\left(\frac{1}{\sqrt{12}}\right)$ & $x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{6}\right)$

4. Find the value of θ, which satisfy $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$.

समीकरण $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$ को हल कीजिए।

Ans. $\theta = 2n\pi$ or $2n\pi + \frac{\pi}{2}$; $n \in I$

Sol. $3 - 2\cos\theta - 4\sin\theta - \cos 2\theta + \sin 2\theta = 0$

$\Rightarrow 3 - 2\cos\theta - 4\sin\theta - (-2\sin^2\theta + 1) + 2\sin\theta\cos\theta = 0$

$\Rightarrow 3 + 2\cos\theta(\sin\theta - 1) - 4\sin\theta + 2\sin^2\theta - 1 = 0$

$\Rightarrow 2\sin^2\theta - 4\sin\theta + 2 + 2\cos\theta(\sin\theta - 1) = 0$

$\Rightarrow 2(\sin\theta - 1)^2 + 2\cos\theta(\sin\theta - 1) = 0$

$\Rightarrow (\sin\theta - 1)(\sin\theta + \cos\theta - 1) = 0$

$\Rightarrow \sin\theta = 1$ or $\sin\theta + \cos\theta = 1$

$\Rightarrow \sin\theta = 1$ or $\cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow \theta = (4n+1)\frac{\pi}{2}; n \in I$ or $\theta - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}; n \in I$

$\Rightarrow \theta = 2n\pi + \frac{\pi}{2}; n \in I$ or $\theta = 2n\pi; n \in I$

5. Find the least positive angle measured in degrees satisfying the equation $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$.

न्यूनतम धनात्मक कोण जो समीकरण $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$ को संतुष्ट करता हो, का डिग्री में मान ज्ञात कीजिए।

Ans. 72°

Sol. Given $\sin^3 x + \sin^3 2x + \sin^3 3x = (\sin x + \sin 2x + \sin 3x)^3$

$$\left.\begin{array}{l} \text{If } a^3 + b^3 + c^3 = (a+b+c)^3 \\ \text{then } 3(a+b)(b+c)(c+a) = 0 \end{array}\right\}$$

$\Rightarrow (\sin x + \sin 2x)(\sin 2x + \sin 3x)(\sin 3x + \sin x) = 0$

$\Rightarrow 8 \sin\frac{3x}{2} \cos^2\frac{x}{2} \cdot \sin 3x \cdot \cos x \cdot \sin\frac{5x}{2} = 0$

$\Rightarrow \sin 3x = 0$ or $\cos x = 0$ or $\cos\frac{x}{2} = 0$

or $\sin\frac{3x}{2} = 0$ or $\sin\frac{5x}{2} = 0$

$\Rightarrow x = 0, 3\pi, \frac{\pi}{2}, \frac{3\pi}{2}, \pi, \frac{2\pi}{3}, \frac{2\pi}{5}$

$\Rightarrow$ Smallest positive value of x is $\frac{2\pi}{5} = 72°$

6. Prove that the equations :

(a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$ (b) $\sin x \cdot \cos 4x \cdot \sin 5x = -1/2$

have no solution.

सिद्ध कीजिए की निम्न समीकरणों का कोई हल नहीं होगा :

(a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$ (b) $\sin x \cdot \cos 4x \cdot \sin 5x = -1/2$

Ans.

Sol. (a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$

$\because \sin x \in [-1, 1]$

$\sin 2x \in [-1, 1]$

$\sin 3x \in [-1, 1]$

This is possible only when

(i) $\sin x = 1$ & $\sin 2x = 1$ & $\sin 3x = 1$

This is not possible.

(ii) $\sin x = -1$ & $\sin 2x = 1$ & $\sin 3x = -1$

This is not possible.

$\Rightarrow \sin x \cdot \sin 2x \cdot \sin 3x = 1$

Equation has no solution.

(b) $\sin x \cdot \cos 4x \cdot \sin 5x = -\frac{1}{2}$

$\Rightarrow (2 \sin x \sin 5x)(\cos 4x) = -1$

$\Rightarrow (\cos 4x - \cos 6x)\cos 4x = -1$

$\Rightarrow \cos^2 4x + 1 = \cos 6x \cos 4x$

$\Rightarrow 2(\cos^2 4x + 1) = 2\cos 6x \cos 4x = \cos 10x + \cos 2x$

$\cos 10x + \cos 2x \in [-2, 2]$ & $2(\cos^2 x + 1) \in [2, 4]$

$\therefore$ Only possibility are $\cos^2 4x + 1 = 1$ & $\cos 10x + \cos 2x = 2$

$\Rightarrow \cos^2 4x = 0$ & $\cos 10x = 1$ & $\cos 2x = 1$

$\because \cos^2 2x = \frac{1+\cos 4x}{2} = \frac{1}{2} \Rightarrow \cos 2x = \frac{\pm 1}{\sqrt{2}}$

$\therefore \sin x \cdot \cos 4x \cdot \sin 5x = \frac{-1}{2}$ have no solution.

7. Solve the equation : $\sin 5x = 16 \sin^5 x$.

समीकरण : $\sin 5x = 16 \sin^5 x$ को हल करें।

Ans. $x = n\pi$ or $x = n\pi \pm \frac{\pi}{6}$

Sol. $\sin 5x = 16 \sin^5 x$

$\Rightarrow \sin(3x + 2x) = 16 \sin^5 x$

$\Rightarrow \sin 3x \cos 2x + \cos 3x \sin 2x = 16 \sin^5 x$

$\Rightarrow (3 \sin x - 4\sin^3 x)(1 - 2\sin^2 x) + (4\cos^3 x - 3\cos x) 2 \sin x \cos x = 16 \sin^5 x$

$\Rightarrow 3 \sin x - 10 \sin^3 x + 8 \sin^5 x + 8 \sin x \cos^4 x - 6 \sin x \cos^2 x = 16 \sin^5 x$

$\Rightarrow 3 \sin x - 10 \sin^3 x + 8 \sin^5 x - 6 \sin x + 8 \sin x + 8 \sin^5 x - 16 \sin^3 x + 6 \sin^3 x = 16 \sin^5 x$

$\Rightarrow 5 \sin x - 20 \sin^3 x = 0$

$\Rightarrow \sin x(1 - 4\sin^2 x) = 0$

$\Rightarrow \sin x = 0$ or $\sin^2 x = \frac{1}{4} = \sin^2 \frac{\pi}{6}$

$\Rightarrow x = n\pi;\ n \in I$ or $x = n\pi \pm \frac{\pi}{6};\ n \in I$

8. Find all the solutions of, $4\cos^2x \sin x - 2\sin^2x = 3\sin x$.

$4\cos^2x \sin x - 2\sin^2x = 3\sin x$ के सभी हल निकालिए।

Ans. $n\pi$; $n\pi + (-1)^n \frac{\pi}{10}$ or $n\pi + (-1)^n \left(-\frac{3\pi}{10}\right)$

Sol. $4\cos^2x \sin x - 2\sin^2x = 3\sin x$

$\Rightarrow 4(1 - \sin^2x)\sin x - 2\sin^2x = 3\sin x$

$\Rightarrow 4\sin x - 4\sin^3x - 2\sin^2x - 3\sin x = 0$

$\Rightarrow \sin x(1 - 2\sin x - 4\sin^2x) = 0$

$\Rightarrow \sin x = 0$ or $4\sin^2x + 2\sin x - 1 = 0$

$\Rightarrow \sin x = 0$ or $\sin x = \frac{\sqrt{5}-1}{4}$ or $\sin x = \frac{-\left(-1+\sqrt{5}\right)}{4}$

$\Rightarrow x = n\pi; n \in I$ or $x = n\pi + (-1)^n \frac{\pi}{10}; n \in I$ or $x = n\pi + (-1)^n \left(-\frac{3\pi}{10}\right)$

9. Solve for x, $(-\pi \le x \le \pi)$ the equation; $2(\cos x + \cos 2x) + \sin 2x(1 + 2\cos x) = 2\sin x$.

x, $(-\pi \le x \le \pi)$ के मानों के लिये समीकरण $2(\cos x + \cos 2x) + \sin 2x(1 + 2\cos x) = 2\sin x$ को हल कीजिए।

Ans. $\frac{\pm\pi}{3}, \frac{-\pi}{2}, \pm\pi$

Sol. $2(\cos x + \cos 2x) + \sin 2x(1 + 2\cos x) = 2\sin x$

$\Rightarrow 2(\cos x + 1 - 2\sin^2x) + 2\sin x\cos x(1 + 2\cos x) = 2\sin x$

$\Rightarrow 2\cos x + 2 - 4\sin^2x + 2\sin x\cos x + 4\sin x\cos^2x - 2\sin x = 0$

$\Rightarrow 2\cos x + 2\sin x\cos x - 2\sin x + 2 - 4\sin^2x + 4\sin x - 4\sin^3x = 0$

$\Rightarrow 2\cos x + 2\sin x(1 + \cos x) + 2 - 4\sin^2x - 4\sin^3x = 0$

$\Rightarrow 2\cos x(1 + \sin x) + 2(\sin x + 1) - 4\sin^2x(1 + \sin x) = 0$

$\Rightarrow (1 + \sin x)(2\cos x + 2 - 4\sin^2x) = 0$

$\Rightarrow (\sin x + 1)(\cos x + 1 - 2(1 - \cos^2x)) = 0$

$\Rightarrow (\sin x + 1)(2\cos^2x + \cos x - 1) = 0$

$\Rightarrow \sin x = -1$ or $\cos x = -1$ or $\cos x = \frac{1}{2}$

$\Rightarrow \sin x = -1$ or $\cos x = -1$ or $\cos x = -\frac{1}{2}$

$\Rightarrow x = \pm\frac{\pi}{3}, \frac{-\pi}{2}, \pm\pi$

10. Find the general solution of the equation $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$.

समीकरण $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$ का व्यापक हल बताइए।

Ans. $x = 2n\pi$ or $x = n\pi + (-1)^n\left(-\frac{\pi}{2}\right)$ or $x = n\pi + (-1)^n\frac{\pi}{6}$

Sol. $2(\sin x - \cos 2x) - \sin 2x(1 + 2\sin x) + 2\cos x = 0$

$\Rightarrow 2\sin x - 2(1 - 2\sin^2 x) - 2\sin x\cos x(1 + 2\sin x) + 2\cos x = 0$

$\Rightarrow 2\sin x - 2 + 4\sin^2 x - 2\sin x\cos x - 4\sin^2 x\cos x + 2\cos x = 0$

$\Rightarrow 2\sin x(1 - \cos x) + 4\sin^2 x(1 - \cos x) - 2(1 - \cos x) = 0$

$\Rightarrow (1 - \cos x)(2\sin x + 4\sin^2 x - 2) = 0$

$\Rightarrow (1 - \cos x)(\sin x + 2\sin^2 x - 1) = 0$

$\Rightarrow \cos x = 1$ or $\sin x = -1$ or $\sin x = \frac{1}{2}$

$\Rightarrow x = 2n\pi;\ n \in I$ or $x = n\pi + (-1)^n\left(-\frac{\pi}{2}\right)$ or $x = n\pi + (-1)^n\frac{\pi}{6}$

11. Find the values of x, between 0 and 2π, satisfying the equation $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$.

0 से 2π में समीकरण $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$ को संतुष्ट करने वाले x के मान ज्ञात कीजिए।

Ans. $\frac{\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{13\pi}{7}$

Sol. $\cos 3x + \cos 2x = \sin\frac{3x}{2} + \sin\frac{x}{2}$

$\Rightarrow \cos\frac{5x}{2}\cos\frac{x}{2} = \sin(x)\cos\frac{x}{2}$

$\Rightarrow \cos\frac{x}{2}\left(\cos\frac{5x}{2} - \sin x\right) = 0$

$\Rightarrow \cos\frac{x}{2} = 0$ or $\cos\left(\frac{5x}{2}\right) = \cos\left(\frac{\pi}{2} - x\right)$

$\Rightarrow \cos\frac{x}{2} = 0$ or $\frac{5x}{2} = 2n\pi \pm \frac{\pi}{2} - x$

$\because\ 0 < x < 2\pi$

$\Rightarrow \frac{x}{2} = \frac{\pi}{2}$ or $x = \frac{\pi}{7}, \frac{9\pi}{7}, \frac{13\pi}{7}, \pi, \frac{5\pi}{7}$

$\Rightarrow x = \frac{\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{13\pi}{7}$

12. Solve: $\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$.

$\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$ को हल करें।

Ans. $x = \frac{n\pi}{4} + (-1)^n \frac{\pi}{8}$ or $\frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$

Sol. $\tan^2 2x + \cot^2 2x + 2\tan 2x + 2\cot 2x = 6$

$\Rightarrow (\tan 2x + \cot 2x)^2 + 2(\tan 2x + \cot 2x) = 8$

Let $\tan 2x + \cot 2x = t$

$\Rightarrow t^2 + 2t - 8 = 0$

$\Rightarrow t = 2$ or $t = -4$

$\Rightarrow \tan^2 2x - 2\tan 2x + 1 = 0$ or $\tan^2 2x + 4\tan 2x + 1 = 0$

$\Rightarrow \tan 2x = 1$ or $\tan^2 2x + 4\tan 2x + 1 = 0$

$\Rightarrow 2x = n\pi + \frac{\pi}{4}$ or $\frac{\sin^2 2x}{\cos^2 2x} + 1 = \frac{-4\sin 2x}{\cos 2x}$

$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}; n \in I$ or $\frac{1}{\cos^2 2x} = \frac{-4\sin 2x}{\cos 2x}$

or $\sin 4x = \frac{-1}{2} = \sin\left(-\frac{\pi}{6}\right)$

$\Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}; n \in I$ or $x = \frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$

13. Solve the equation: $1 + 2\operatorname{cosec} x = -\frac{\sec^2 \frac{x}{2}}{2}$.

$1 + 2\operatorname{cosec} x = -\frac{\sec^2 \frac{x}{2}}{2}$ को हल करें।

Ans. $x = 2n\pi - \frac{\pi}{2}$

Sol. Let $\tan \frac{x}{2} = t$

$\Rightarrow \sin x = \frac{2t}{1+t^2}$ & $\sec^2 \frac{x}{2} = 1 + t^2$

$\Rightarrow 1 + \frac{2(1+t^2)}{2t} + \frac{1}{2}(1+t^2) = 0$

$$\Rightarrow 1+\frac{1+t^2}{t}+\frac{1}{2}+\frac{t^2}{2}=0$$

$\Rightarrow t^3 + t + 2 + 2t^2 + 2t = 0$

$\Rightarrow t^3 + 2t^2 + 3t + 2 = 0$

$\Rightarrow (t + 1)(t^2 + t + 2) = 0$

$\Rightarrow t = -1$

$$\Rightarrow \tan\frac{x}{2}=\tan\left(-\frac{\pi}{4}\right)$$

$$\Rightarrow \frac{x}{2}=n\pi-\frac{\pi}{4}$$

$$\Rightarrow x=2n\pi-\frac{\pi}{2};n\in I$$

14. Solve: $\tan^2x \cdot \tan^2 3x \cdot \tan 4x = \tan^2x - \tan^2 3x + \tan 4x$.

$\tan^2x \cdot \tan^2 3x \cdot \tan 4x = \tan^2x - \tan^2 3x + \tan 4x$ को हल करें।

Ans. $\frac{(2n+1)\pi}{4}$, $k\pi$, where $n, k \in I$

Sol. $\tan 4x\,[(\tan x \tan 3x)^2 - 1] = \tan^2x - \tan^2 3x$

$$\Rightarrow \tan 4x=\frac{\tan^2 x-\tan^2 3x}{\tan^2 x\tan^2 3x-1}=\frac{\tan^2 3x-\tan^2 x}{1-\tan^2 x\tan^2 3x}=\tan(3x-x)\cdot\tan(3x+x)$$

$\Rightarrow \tan 4x = \tan 2x \tan 4x$

$\Rightarrow \tan 4x = 0$ or $\tan 2x = 1$ (Not possible as tan4x is not defined)

$$\Rightarrow x=\frac{n\pi}{4}$$

$$\Rightarrow x=\frac{n\pi}{4};n\in I$$

$$\because x\neq(2n+1)\frac{\pi}{2}$$

$$\Rightarrow x=\frac{(2n+1)\pi}{4};n\in I \quad \text{or} \quad x=k\pi;\ k\in I$$

15. Find the set of values of x satisfying the equality $\sin\left(x-\frac{\pi}{4}\right) - \cos\left(x+\frac{3\pi}{4}\right) = 1$ and the inequality

$$\frac{2\cos 7x}{\cos 3+\sin 3}>2^{\cos 2x}.$$

समीकरण $\sin\left(x-\frac{\pi}{4}\right)-\cos\left(x+\frac{3\pi}{4}\right)=1$ तथा असमिका $\frac{2\cos 7x}{\cos 3+\sin 3}>2^{\cos 2x}$ को सन्तुष्ट करने वाले x के मान ज्ञात करें।

Ans. $x=2n\pi+\frac{3\pi}{4}$, $n\in I$

Sol. $\sin\left(x-\frac{\pi}{4}\right)-\cos\left(x+\frac{3\pi}{4}\right)=1$

$\Rightarrow \sin x\cos\frac{\pi}{4}-\cos x\cos\frac{\pi}{4}-\cos x\cos\frac{3\pi}{4}+\sin x\sin\frac{3\pi}{4}=1$

$\Rightarrow \sqrt{2}\sin x=1$

$\Rightarrow \sin x=\frac{1}{\sqrt{2}}\Rightarrow x=\frac{\pi}{4},\frac{3\pi}{4}$

$\Rightarrow \cos 2x=1-2\sin^2 x=1-\frac{2}{2}=1-1=0$

$\Rightarrow \frac{2\cos 7x}{\cos x3+\sin 3}>2^{\cos 2x}$

$\Rightarrow \frac{2\cos 7x}{\cos 3+\sin 3}>1$

$\because \cos 3+\sin 3<0$

$\Rightarrow \cos 7x<0$ If $x=\frac{3\pi}{4}$

$\therefore$ equation has solution.

$\Rightarrow$ General solution is $x=2n\pi+\frac{3\pi}{4};n\in I$

16. Solve: $\sin\left(\frac{\sqrt{x}}{2}\right)+\cos\left(\frac{\sqrt{x}}{2}\right)=\sqrt{2}\sin\sqrt{x}$.

$\sin\left(\frac{\sqrt{x}}{2}\right)+\cos\left(\frac{\sqrt{x}}{2}\right)=\sqrt{2}\sin\sqrt{x}$ को हल करें।

Ans. $x=\left(4n\pi+\frac{\pi}{2}\right)^2$ or $x=\left(\frac{4m\pi}{3}+\frac{\pi}{2}\right)^2$ where $m, n\in W$.

Sol. $\sin\left(\frac{\sqrt{x}}{2}\right)+\cos\left(\frac{\sqrt{x}}{2}\right)=\sqrt{2}\sin\sqrt{x}$

$\Rightarrow \frac{1}{\sqrt{2}}\sin\left(\frac{\sqrt{x}}{2}\right)+\frac{1}{\sqrt{2}}\cos\left(\frac{\sqrt{x}}{2}\right)=\sin\sqrt{x}$

For $\sqrt{x}$; $x \geq 0$

$\Rightarrow \sin\left(\frac{\pi}{4}+\frac{\sqrt{x}}{2}\right)=\sin\sqrt{x}$

$\Rightarrow \sqrt{x}=n\pi+(-1)^n\left(\frac{\pi}{4}+\frac{\sqrt{x}}{2}\right)$

$\Rightarrow \sqrt{x}-\frac{\sqrt{x}}{2}=2n\pi+\frac{\pi}{4}$ or $\sqrt{x}+\frac{\sqrt{x}}{2}=(2m+1)\pi-\frac{\pi}{4}$

$\Rightarrow \sqrt{x}=4n\pi+\frac{\pi}{2}$ or $\sqrt{x}=\frac{4m\pi}{3}+\frac{\pi}{2}$

$\Rightarrow x=\left(4n\pi+\frac{\pi}{2}\right)^2 ; n \in W$ or $x=\left(\frac{4m\pi}{3}+\frac{\pi}{2}\right)^2 ; m \in W$

17. Solve for x and y, $\begin{matrix} x\cos^3 y + 3x\cos y \sin^2 y = 14 \\ x\sin^3 y + 3x\cos^2 y \sin y = 13 \end{matrix}$.

समीकरणों $\begin{matrix} x\cos^3 y + 3x\cos y \sin^2 y = 14 \\ x\sin^3 y + 3x\cos^2 y \sin y = 13 \end{matrix}$ के x, y के लिये हल करें।

Ans. $x = \pm 5\sqrt{5}$ & $y = n\pi + \tan^{-1}\frac{1}{2}$

Sol. $x\cos^3 y + 3x\cos y \sin^2 y = 14$ ...(i)

$x\sin^3 y + 3x\cos^2 y \sin y = 13$ ...(ii)

By (i) + (ii); we get

$x(\cos^3 y + \sin^3 y) + 3x \sin y \cos y (\sin y + \cos y) = 27$

$\Rightarrow x((\cos y + \sin y)^3 - 3\cos y \sin y (\cos y + \sin y)) + 3x \sin y \cos y (\sin y + \cos y) = 27$

$\Rightarrow x(\cos y + \sin y)^3 = 27$

$\Rightarrow x = \left(\frac{3}{\cos y + \sin y}\right)^3$ ...(iii)

By (i) – (ii); we get

$x(\cos^3 y - \sin^3 y) + 3x \sin y \cos y (\sin y - \cos y) = 1$

$\Rightarrow x(\cos y - \sin y)^3 = 1$

$$\Rightarrow x = \frac{1}{(\cos y - \sin y)^3} \quad \text{...(iv)}$$

By (iii) & (iv)

$$\frac{\cos y + \sin y}{\cos y - \sin y} = \frac{3}{1}$$

Apply C & D Rule

$$\tan y = \frac{1}{2}$$

$$\Rightarrow y = n\pi + \tan^{-1}\left(\frac{1}{2}\right); n \in I$$

$$\Rightarrow x = \pm 5\sqrt{5}$$

18. Let $A = \left\{\theta \in R \,\middle|\, \cos^2(\sin\theta) + \sin^2(\cos\theta) = 1\right\}$ and $B = \left\{\theta \in R \,\middle|\, \cos(\sin\theta) \cdot \sin(\cos\theta) = 0\right\}$, then find the number of elements in the set $A \cap B$.

माना $A = \left\{\theta \in R \,\middle|\, \cos^2(\sin\theta) + \sin^2(\cos\theta) = 1\right\}$ तथा $B = \left\{\theta \in R \,\middle|\, \cos(\sin\theta) \cdot \sin(\cos\theta) = 0\right\}$ है, तो समुच्चय $A \cap B$ में अवयवों की संख्या ज्ञात कीजिये।

Ans. 0

Sol. $A = \{\theta \in R \mid \cos^2(\sin\theta) + \sin^2(\cos\theta) = 1\}$

$B = \{\theta \in R \mid \cos(\sin\theta) \cdot \sin(\cos\theta) = 0\}$

If $\theta \in A \cap B$

then

$\cos^2(\sin\theta) + \sin^2(\cos\theta) = 1$ & $\cos(\sin\theta) \cdot \sin(\cos\theta) = 0$

If $\cos(\sin\theta) = 0$

Then $\underbrace{\sin\theta = (2n+1)\frac{\pi}{2}; n \in I}_{\text{Not possible}}$

If $\sin(\cos\theta) = 0$ then $\cos\theta = n\pi$; $n \in I$

Only possibility are $\cos\theta = 0$ but $\cos^2(1) \neq 1$

$\therefore$ No. of element in $A \cap B$ is 0.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 19 to 21

Let S_1 be the set of all those solutions of the equation $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$ which are independent of a and b and S_2 be the set of all such solutions which are dependent on a and b. Then

माना S_1 समीकरण $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$ के उन सभी हलों का समुच्चय है जो a तथा b से स्वतंत्र हैं तथा S_2 उन सभी हलों का समुच्चय है जो a तथा b पर निर्भर करते हैं।

19. The set S_1 and S_2 are :

(A) $\{n\pi;\ n \in I\}$ and $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \frac{\sin^{-1}(a\sin b)}{2};\ n \in I\right\}$

(B) $\{n\pi;\ n \in I\}$ and $\left\{\frac{n\pi}{2} + (-1)^n \frac{\sin^{-1}(b\sin a)}{2};\ n \in I\right\}$

(C) $\left\{\frac{n\pi}{2};\ \ n \in I\right\}$ and $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \sin^{-1}\left(\frac{a}{2}\sin b\right);\ n \in I\right\}$

(D) none of these

समुच्चय S_1 तथा S_2 है :

(A) $\{n\pi;\ n \in I\}$ तथा $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \frac{\sin^{-1}(a\sin b)}{2};\ n \in I\right\}$

(B) $\{n\pi;\ n \in I\}$ तथा $\left\{\frac{n\pi}{2} + (-1)^n \frac{\sin^{-1}(b\sin a)}{2};\ n \in I\right\}$

(C) $\left\{\frac{n\pi}{2};\ \ n \in I\right\}$ तथा $\left\{\frac{n\pi}{2} + \frac{b}{2} + (-1)^n \sin^{-1}\left(\frac{a}{2}\sin b\right);\ n \in I\right\}$

(D) इनमें से कोई नहीं

Ans. A

Sol. $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$

$\Rightarrow \cos\theta\cos(2\theta - b) + a\cos\theta\cos(2\theta - b) = \cos(\theta - b) + a\cos 2\theta\cos(\theta - b)$

$\Rightarrow 2\cos\theta\cos(2\theta - b) + 2a\cos\theta\cos(2\theta - b)$

$= 2\cos(\theta - b) + 2a\cos 2\theta\cos(\theta - b)$

$\Rightarrow \cos(3\theta - b) + \cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta - b))$

$= 2\cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta + b))$

$\Rightarrow \cos(3\theta - b) + a\cos(\theta - b) = \cos(\theta - b) + a\cos(\theta + b)$

$\Rightarrow \cos(3\theta - b) - \cos(\theta - b) = a(\cos(\theta + b) - \cos(\theta - b))$

$\Rightarrow$ $2\sin(2\theta - b)\sin\theta = 2a\sin\theta\sin b$

$\Rightarrow$ $\sin\theta(\sin(2\theta - b) - a\sin b) = 0$

$\Rightarrow$ $\sin\theta = 0$ or $\sin(2\theta - b) = a\sin b$

$\Rightarrow$ $\sin\theta = 0$ $\Rightarrow \theta = n\pi$, $n \in I$.

and $\sin(2\theta - b) = a\sin b$.

$\Rightarrow$ $\sin(2\theta - b) = \sin(\sin^{-1}(a\sin b))$

$\Rightarrow$ $2\theta - b = n\pi + (-1)^n \sin^{-1}(a\sin b)$ $n \in I$

$S_1 = n\pi$, $n \in I$.

$S_2 = \frac{1}{2}(n\pi + (-1)^n \sin^{-1}(a\sin b) + b)$ $n \in I$.

20. Conditions that should be imposed on a and b such that S_2 is non-empty :

(A) $\left|\frac{a}{2}\sin b\right| < 1$ (B) $\left|\frac{a}{2}\sin b\right| \le 1$ (C) $|a\sin b| \le 1$ (D) none of these

a तथा b पर शर्त (condition) लगानी चाहिये ताकि S_2 अरिक्त हो :

(A) $\left|\frac{a}{2}\sin b\right| < 1$ (B) $\left|\frac{a}{2}\sin b\right| \le 1$ (C) $|a\sin b| \le 1$ (D) इनमें से कोई नहीं

Ans. C

Sol. $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$

$\Rightarrow$ $\cos\theta\cos(2\theta - b) + a\cos\theta\cos(2\theta - b) = \cos(\theta - b) + a\cos 2\theta\cos(\theta - b)$

$\Rightarrow$ $2\cos\theta\cos(2\theta - b) + 2a\cos\theta\cos(2\theta - b)$

$= 2\cos(\theta - b) + 2a\cos 2\theta\cos(\theta - b)$

$\Rightarrow$ $\cos(3\theta - b) + \cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta - b))$

$= 2\cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta + b))$

$\Rightarrow$ $\cos(3\theta - b) + a\cos(\theta - b) = \cos(\theta - b) + a\cos(\theta + b)$

$\Rightarrow$ $\cos(3\theta - b) - \cos(\theta - b) = a(\cos(\theta + b) - \cos(\theta - b))$

$\Rightarrow$ $2\sin(2\theta - b)\sin\theta = 2a\sin\theta\sin b$

$\Rightarrow$ $\sin\theta(\sin(2\theta - b) - a\sin b) = 0$

$\Rightarrow$ $\sin\theta = 0$ or $\sin(2\theta - b) = a\sin b$

$\Rightarrow$ $\sin\theta = 0$ $\Rightarrow \theta = n\pi$, $n \in I$.

and $\sin(2\theta - b) = a\sin b$.

$\Rightarrow$ $\sin(2\theta - b) = \sin(\sin^{-1}(a\sin b))$

$\Rightarrow$ $2\theta - b = n\pi + (-1)^n \sin^{-1}(a\sin b)$ $n \in I$

$|a\sin b| \le 1 \rightarrow$ For S_2 non empty.

21. All the permissible values of b, if $a = 0$ and $S_2 \cap (0, \pi)$ is non empty, is :

(A) $b \in (-n\pi, 2n\pi)$, $n \in I$ (B) $b \in (-n\pi, 2\pi - n\pi)$, $n \in I$

(C) $b \in (-n\pi, n\pi)$ (D) none of these

b के सभी अनुमेय मान, यदि $a = 0$ तथा $S_2 \cap (0, \pi)$ अरिक्त हो, होंगे :

(A) $b \in (-n\pi, 2n\pi)$, $n \in I$ (B) $b \in (-n\pi, 2\pi - n\pi)$, $n \in I$

(C) $b \in (-n\pi, n\pi)$ (D) इनमें से कोई नहीं

Ans. B

Sol. $(1 + a)\cos\theta \cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$

$\Rightarrow \cos\theta \cos(2\theta - b) + a\cos\theta\cos(2\theta - b) = \cos(\theta - b) + a\cos 2\theta \cos(\theta - b)$

$\Rightarrow 2\cos\theta\cos(2\theta - b) + 2a\cos\theta\cos(2\theta - b)$

$= 2\cos(\theta - b) + 2a\cos 2\theta\cos(\theta - b)$

$\Rightarrow \cos(3\theta - b) + \cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta - b))$

$= 2\cos(\theta - b) + a(\cos(3\theta - b) + \cos(\theta + b))$

$\Rightarrow \cos(3\theta - b) + a\cos(\theta - b) = \cos(\theta - b) + a\cos(\theta + b)$

$\Rightarrow \cos(3\theta - b) - \cos(\theta - b) = a(\cos(\theta + b) - \cos(\theta - b))$

$\Rightarrow 2\sin(2\theta - b)\sin\theta = 2a\sin\theta\sin b$

$\Rightarrow \sin\theta(\sin(2\theta - b) - a\sin b) = 0$

$\Rightarrow \sin\theta = 0$ or $\sin(2\theta - b) = a\sin b$

$\Rightarrow \sin\theta = 0 \Rightarrow \theta = n\pi, \quad n \in I.$

and $\sin(2\theta - b) = a\sin b.$

$\Rightarrow \sin(2\theta - b) = \sin(\sin^{-1}(a\sin b))$

$\Rightarrow 2\theta - b = n\pi + (-1)^n \sin^{-1}(a\sin b)$ $n \in I$

If $a = 0$

$\sin(2\theta - b) = 0$

$2\theta - b = n\pi$ $n \in I$

for S_2 a subset of $(0,\pi)$

$$0 < \frac{n\pi + b}{2} < \pi \qquad n \in I$$

$\Rightarrow -n\pi < b < 2\pi - n\pi.$

$b \in (-n\pi, 2\pi - n\pi)$, $n \in I.$

MATCH THE COLUMN TYPE QUESTION(S)

22. Match the following column :

Column-I (Equation)	**Column-II (General Solution)**
(A) $\frac{1 - \cos 2x}{\sin 2x} = 0$	(P) $(8n - 1)\frac{\pi}{4}$, $n \in I$
(B) $\cos^2 x + 5\cos x = 2\sin^2 x$	(Q) $2n\pi \pm \cos^{-1}\frac{1}{3}$, $n \in I$
(C) $\sin^4 x + \sin^4\left(\frac{x}{2} + \frac{\pi}{8}\right) + \cos^4 x = \frac{1}{2}\sin^2 2x$	(R) No solution
(D) $4\cos^2\theta \cdot \sin\theta - 2\sin^2\theta = 3\sin\theta$	(S) $n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$, $n \in I$

निम्न कॉलम को सुमेलित कीजिए :

Column-I (समीकरण)	**Column-II (व्यापक हल)**
(A) $\frac{1-\cos 2x}{\sin 2x}=0$	(P) $(8n-1)\frac{\pi}{4}, n \in I$
(B) $\cos^2 x + 5\cos x = 2\sin^2 x$	(Q) $2n\pi \pm \cos^{-1}\frac{1}{3}, n \in I$
(C) $\sin^4 x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) + \cos^4 x = \frac{1}{2}\sin^2 2x$	(R) कोई हल नहीं
(D) $4\cos^2\theta \cdot \sin\theta - 2\sin^2\theta = 3\sin\theta$	(S) $n\pi + (-1)^n\left(-\frac{3\pi}{10}\right), n \in I$

Ans. (A) R, (B) Q, (C) P (D) S

Sol. (A) $\frac{1-\cos 2x}{\sin 2x}=0$

$\Rightarrow \frac{2\sin^2 x}{2\sin x\cos x}=0 \qquad \{\because \sin 2x \neq 0\}$

$\Rightarrow \tan x = 0$

$\Rightarrow x = n\pi; n \in I$

$\because \sin 2x \neq 0$

$\Rightarrow$ This equation has no solution.

(B) $\cos^2 x + 5\cos x = 2\sin^2 x$

$\Rightarrow \cos^2 x + 5\cos x - 2(1-\cos^2 x) = 0$

$\Rightarrow 3\cos^2 x + 5\cos x - 2 = 0$

$\Rightarrow \cos x = -2$ (Not possible as $\cos x \in [-1, 1]$) or $\cos x = \frac{1}{3}$

$\Rightarrow x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right); n \in I$

(C) $\sin^4 x + \cos^4 x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) = \frac{1}{2}\sin^2 2x$

$\Rightarrow 1 - \sin^2 2x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) = 0$

$\Rightarrow \cos^2 2x + \sin^4\left(\frac{x}{2}+\frac{\pi}{8}\right) = 0$

$\Rightarrow \cos 2x = 0$ or $\sin\left(\frac{x}{2}+\frac{\pi}{8}\right) = 0$

$\Rightarrow x = n\pi \pm \frac{\pi}{4}$ or $x = 2n\pi - \frac{\pi}{4}; n \in I$

(D) $4\cos^2\theta \sin\theta - 2\sin^2\theta = 3\sin\theta$

$\Rightarrow 4\sin\theta(1 - \sin^2\theta) - 2\sin^2\theta - 3\sin\theta = 0$

$\Rightarrow \sin\theta - 4\sin^3\theta - 2\sin^2\theta = 0$

$\Rightarrow 4\sin^3\theta + 2\sin^2\theta - \sin\theta = 0$

$\Rightarrow \sin\theta(4\sin^2\theta + 2\sin\theta - 1) = 0$

$\Rightarrow \sin\theta = 0$ or $\sin\theta = \frac{\sqrt{5}-1}{4}$ or $\sin\theta = -\left(\frac{1+\sqrt{5}}{4}\right)$

$\Rightarrow \theta = n\pi$ or $\theta = n\pi + (-1)^n \frac{\pi}{10}$ or $\theta = n\pi + (-1)^n\left(-\frac{3\pi}{10}\right)$

23. Match the following column :

Column-I	**Column-II**
(A) $\cos 3x \cdot \cos^3 x + \sin 3x \cdot \sin^3 x = 0$	(P) $n\pi \pm \frac{\pi}{3}$
(B) $\sin 3\alpha = 4\sin\alpha \sin(x+\alpha)\sin(x-\alpha)$ where α is a constant $\neq n\pi$	(Q) $n\pi + \frac{\pi}{4}, n \in I$
(C) $\lvert 2\tan x - 1\rvert + \lvert 2\cot x - 1\rvert = 2$	(R) $\frac{n\pi}{4} + \frac{\pi}{8}, n \in I$
(D) $\sin^{10}x + \cos^{10}x = \frac{29}{16}\cos^4 2x$	(S) $\frac{n\pi}{2} \pm \frac{\pi}{4}$

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $\cos 3x \cdot \cos^3 x + \sin 3x \cdot \sin^3 x = 0$	(P) $n\pi \pm \frac{\pi}{3}$
(B) $\sin 3\alpha = 4\sin\alpha \sin(x+\alpha)\sin(x-\alpha)$ जहाँ α एक नियतांक है $\neq n\pi$	(Q) $n\pi + \frac{\pi}{4}, n \in I$
(C) $\lvert 2\tan x - 1\rvert + \lvert 2\cot x - 1\rvert = 2$	(R) $\frac{n\pi}{4} + \frac{\pi}{8}, n \in I$
(D) $\sin^{10}x + \cos^{10}x = \frac{29}{16}\cos^4 2x$	(S) $\frac{n\pi}{2} \pm \frac{\pi}{4}$

Ans. (A) S; (B) P; (C) Q; (D) R

Sol. (A) $\cos 3x \cos^3 x + \sin 3x \sin^3 x = 0$

$\Rightarrow (4\cos^3 x - 3\cos x)\cos^3 x + (3\sin x - 4\sin^3 x)\sin^3 x = 0$

$\Rightarrow 4(\cos^6 x - \sin^6 x) + 3(\sin^4 x - \cos^4 x) = 0$

$\Rightarrow \cos 2x [4(1 - \sin^2 x \cos^2 x) - 3] = 0$

$\Rightarrow \cos^3 2x = 0$

$\Rightarrow 2x = 2n\pi \pm \frac{\pi}{2}$

$\Rightarrow x = n\pi \pm \frac{\pi}{4}; n \in I$

(B) $\sin 3\alpha = 4 \sin \alpha \sin (x + \alpha) \sin (x - \alpha); a \neq n\pi$

$\because \sin(\alpha) \sin(60^\circ + \alpha) \sin(60^\circ - \alpha) = \frac{1}{4}\sin 3\alpha$

$\therefore x = n\pi \pm \frac{\pi}{3}$

(C) $|2 \tan x - 1) + |2 \cot x - 1| = 2$

$\Rightarrow (2 \tan x - 1)^2 + (2 \cot x - 1)^2 + 2 |(2 \tan x - 1)(2 \cot x - 1)| = 4$

$\Rightarrow 4(\tan^2 x + \cot^2 x) + 2 - 4 \tan x - 4 \cot x + 2 |4 \tan x \cot x - 2(\tan x + \cot x) + 1| = 4$

Let $\tan x + \cot x = t$

$\Rightarrow 4 (t^2 - 2) + 2 - 4t + 2 |5 - 2t| = 4$

$4t^2 - 8 + 2 - 4t + 2|5 - 2t| = 4$

Case-I : If $5 - 2t > 0$ i.e. $t < \frac{5}{2}$

Then

$4t^2 - 8 + 2 - 4t + 10 - 4t - 4 = 0$

$4t^2 - 8t = 0$

$\Rightarrow \underbrace{t = 0}_{\text{Not possible}}$ or $t = 2$

$\Rightarrow x = n\pi + \frac{\pi}{4}; n \in I$

Case-II : If $5 - 2t < 0$ i.e. $t > \frac{5}{2}$

then

$4t^2 - 8 + 2 - 4t - 10 + 4t - 4 = 0$

$\Rightarrow 4t^2 = 20$

$\Rightarrow t^2 = 5$

$\because$ not possible as $t > \frac{5}{2}$

(D) $\sin^{10} x + \cos^{10} x = \frac{29}{16}\cos^4 2x$

$$\Rightarrow \left(\frac{1-\cos 2x}{2}\right)^5 + \left(\frac{1+\cos 2x}{2}\right)^5 = \frac{29}{16}(\cos 2x)^4$$

let $\cos 2x = t$

$$\Rightarrow \left(\frac{1-t}{2}\right)^5 + \left(\frac{1+t}{2}\right)^5 = \frac{29}{16}t^4$$

$\Rightarrow (1-t)^5 + (1+t)^5 = 58t^4$

$\Rightarrow 2 + 20t^2 + 10t^4 - 58t^4 = 0$

$\Rightarrow 1 + 10t^2 - 24t^4 = 0$

$\Rightarrow 24t^4 - 10t^2 - 1 = 0$

$\Rightarrow (2t^2 - 1)(12t^2 + 1) = 0$

$\Rightarrow 2t^2 - 1 = 0$ or $12t^2 + 1 = 0$ (not possible)

$$\Rightarrow \cos^2 2x = \frac{1}{2}$$

$\Rightarrow \cos 4x = 0$

$$\Rightarrow 4x = n\pi + \frac{\pi}{2}$$

$$\Rightarrow x = \frac{n\pi}{4} + \frac{\pi}{8}; n \in I$$